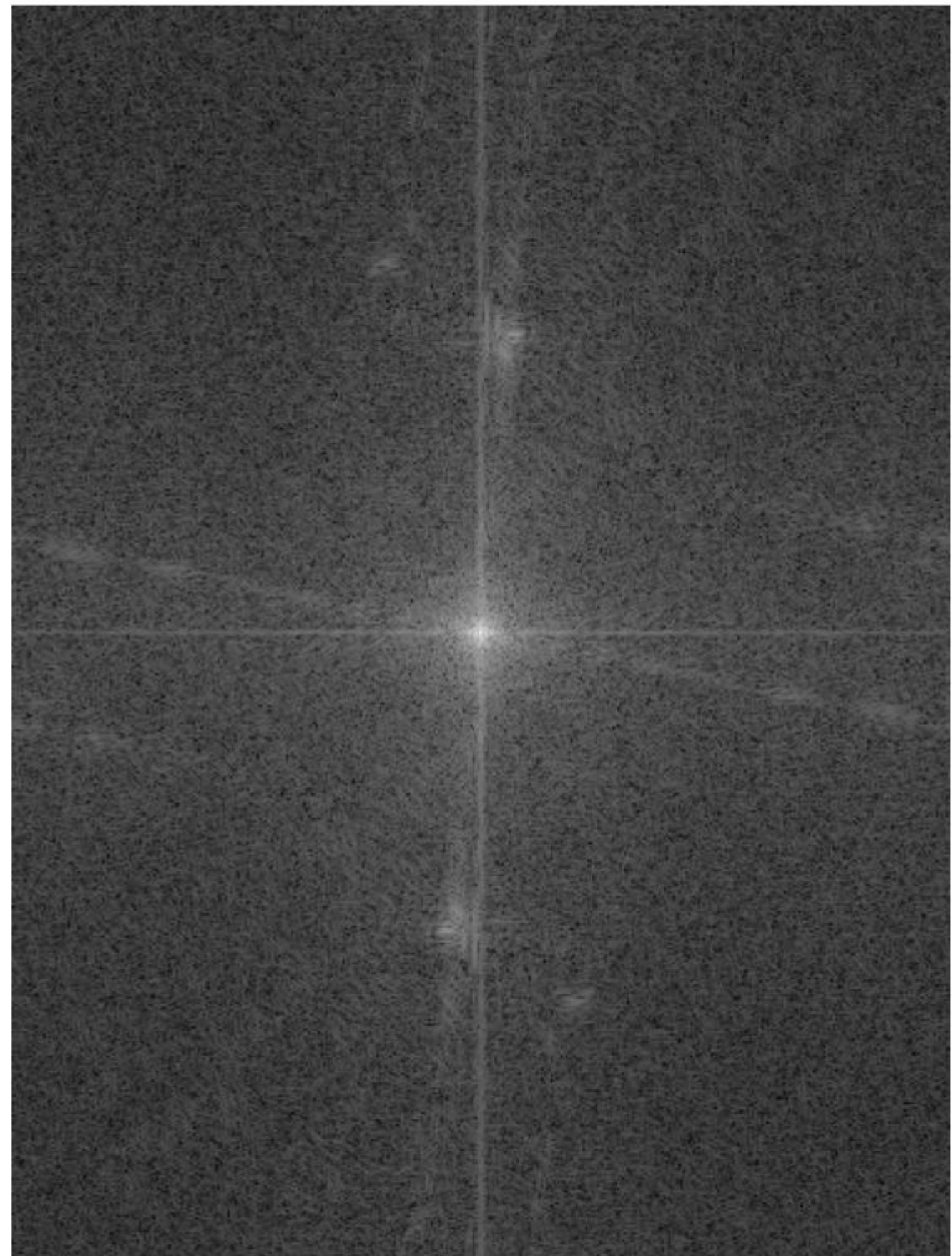


Original (person 0, Neutral)



Magnitude spectrum (log scale)



Original



Ideal, $r=15$, $MSE=0.0086$



Ideal, $r=30$, $MSE=0.0061$



Ideal, $r=60$, $MSE=0.0049$



Ideal, $r=120$, $MSE=0.0040$



Noisy, $MSE=0.0059$



Gaussian, $r=15$, $MSE=0.0073$



Gaussian, $r=30$, $MSE=0.0053$



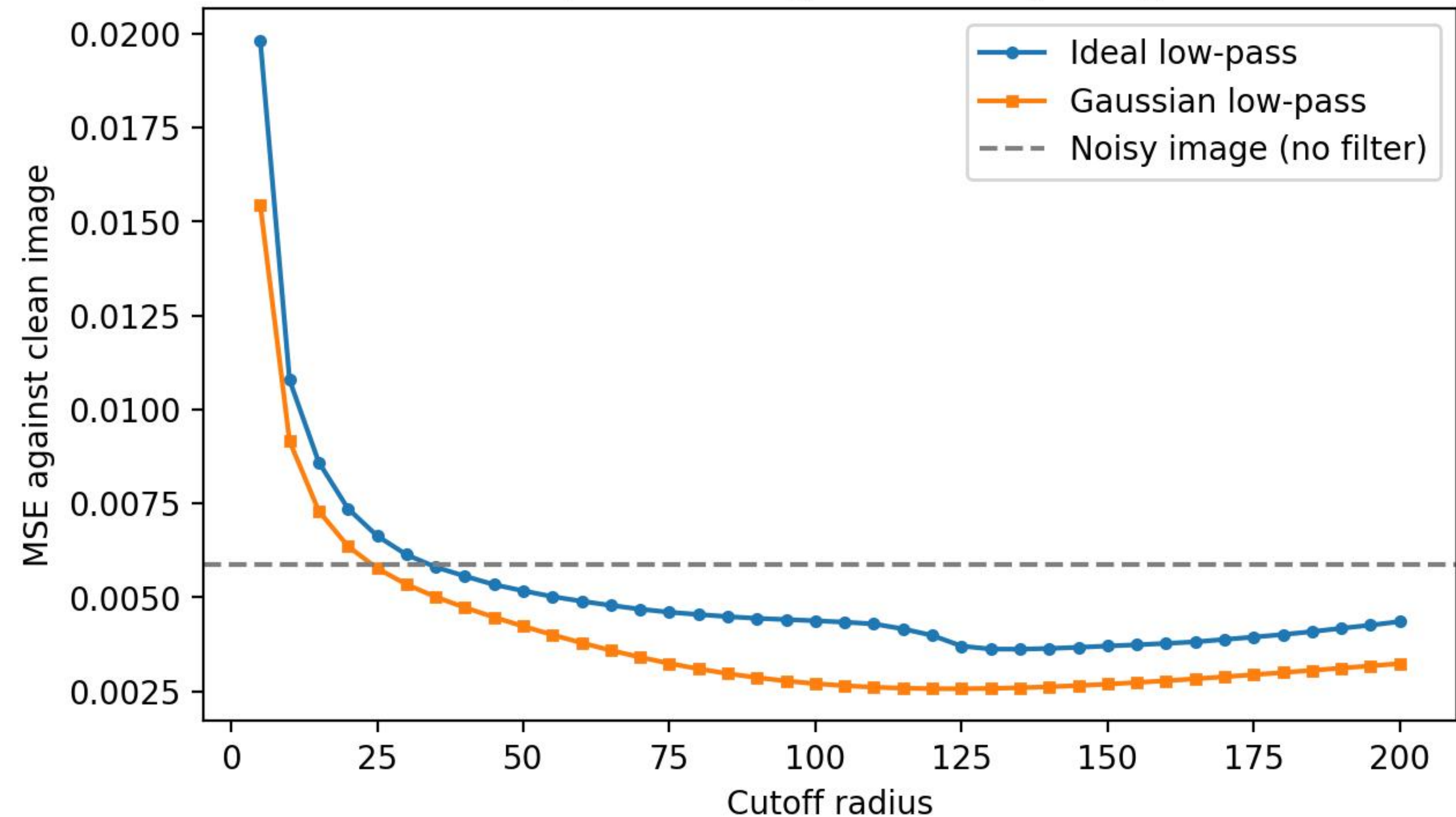
Gaussian, $r=60$, $MSE=0.0038$



Gaussian, $r=120$, $MSE=0.0026$



Low-pass filtering of the noisy image



Original (zoom)



Ideal low-pass, $r=30$ (zoom)



Gaussian low-pass, $r=30$ (zoom)



Original



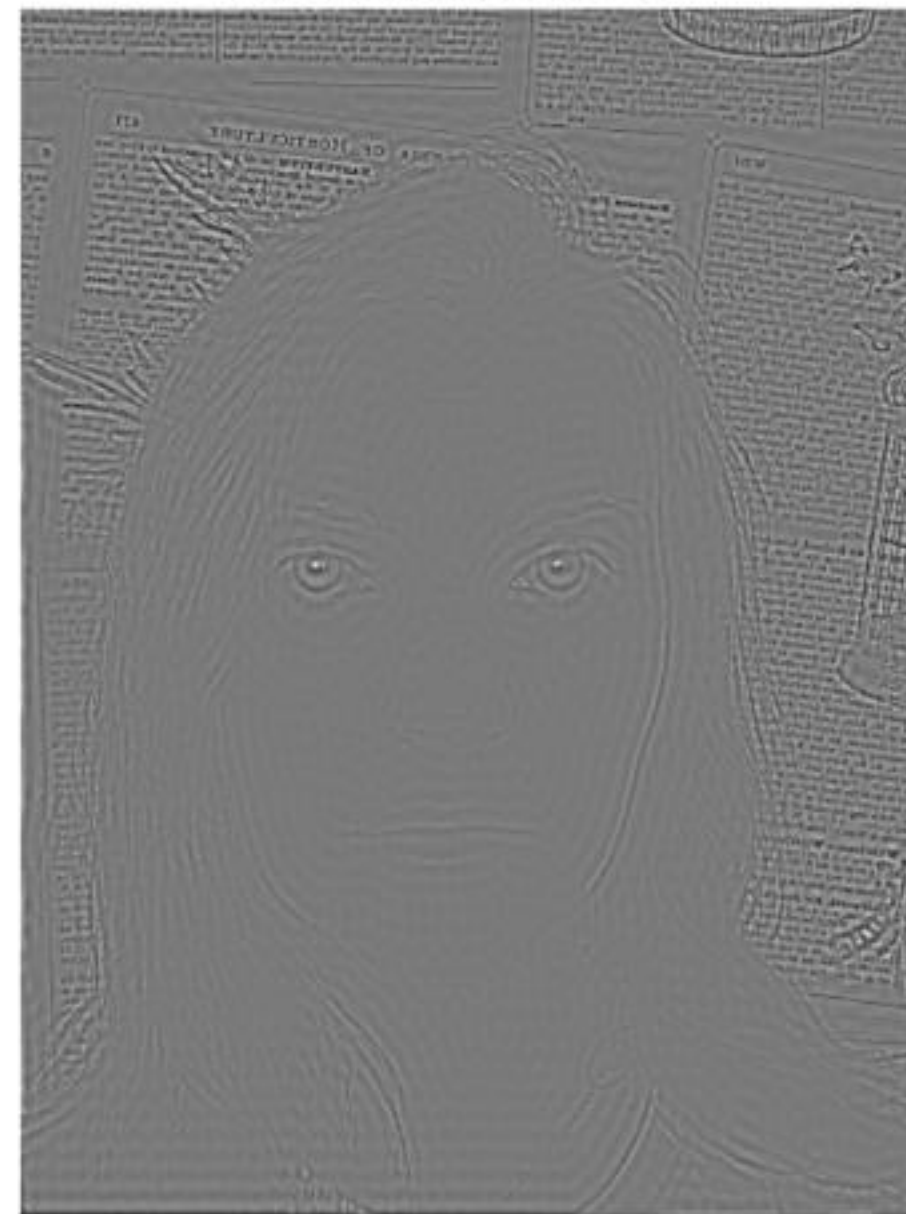
Ideal high-pass, $r=10$



Ideal high-pass, $r=30$



Ideal high-pass, $r=60$



Gaussian high-pass, $r=10$



Gaussian high-pass, $r=30$



Gaussian high-pass, $r=60$



Original



Edge strength |high-pass|, Gaussian $r=30$



Original + high-pass (edge enhanced)



Original



0.5 % kept, PSNR=21.75 dB



1 % kept, PSNR=22.61 dB



2 % kept, PSNR=23.51 dB



5 % kept, PSNR=24.82 dB



10 % kept, PSNR=26.16 dB



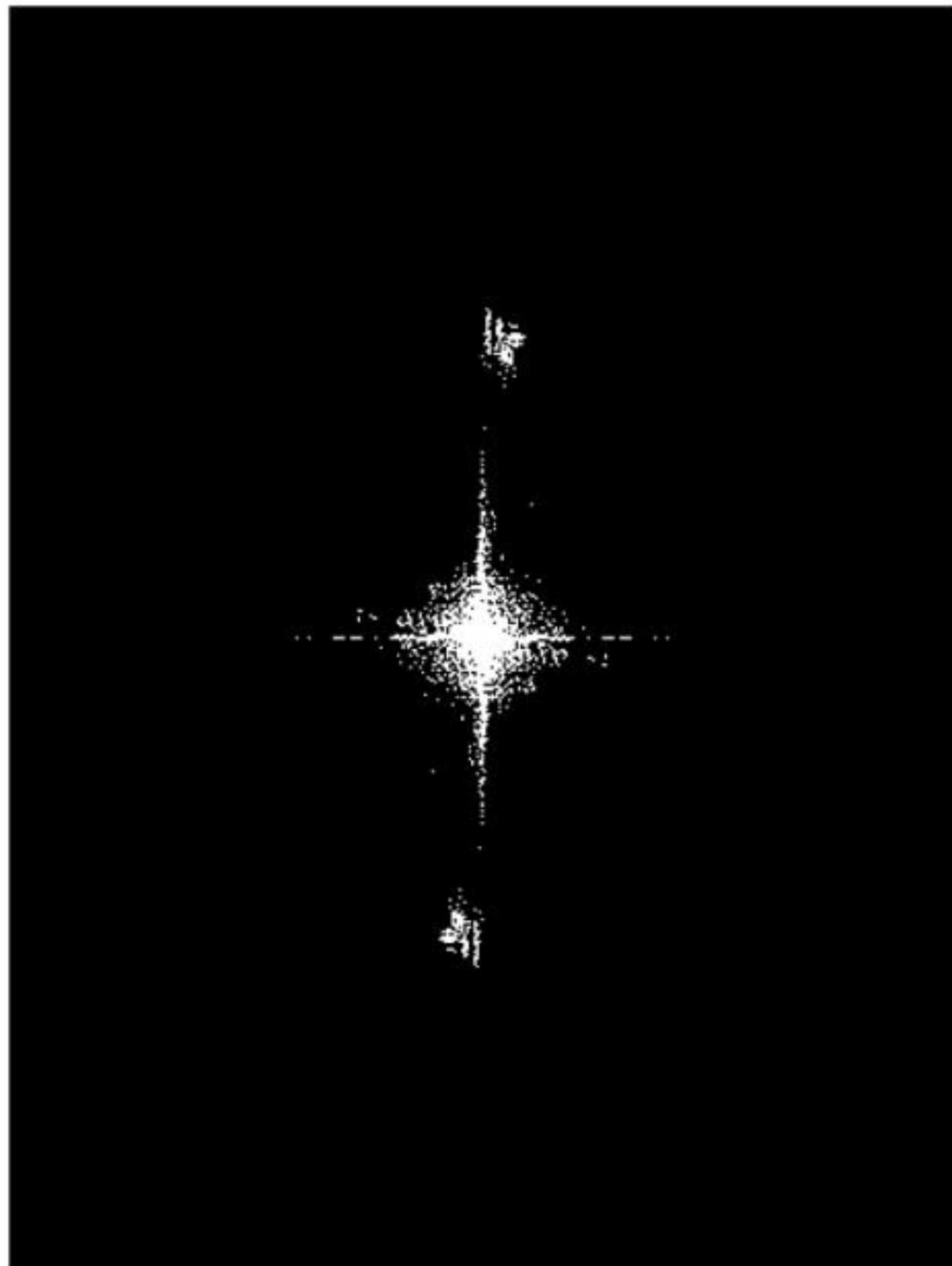
20 % kept, PSNR=28.2 dB



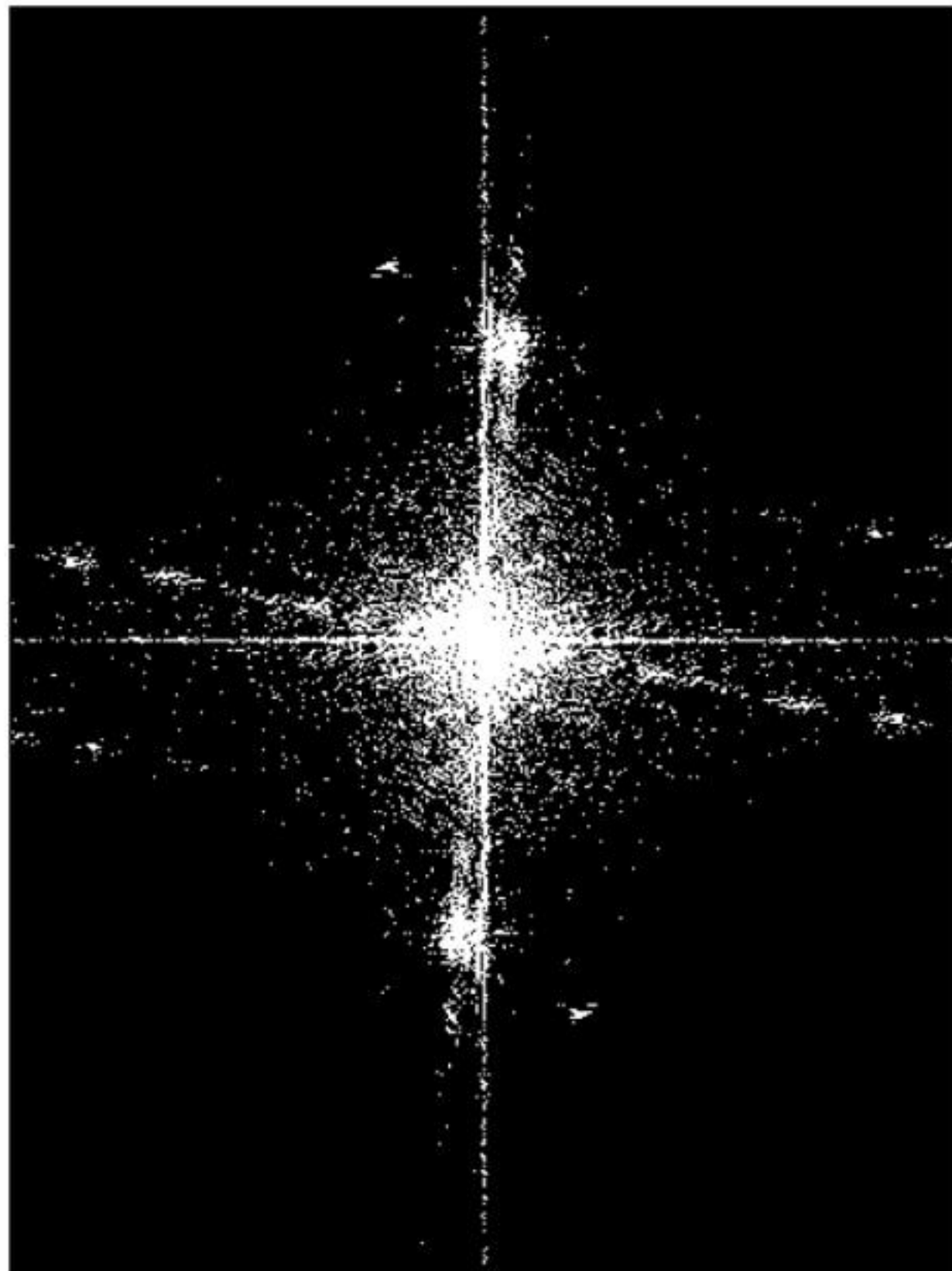
50 % kept, PSNR=33.87 dB



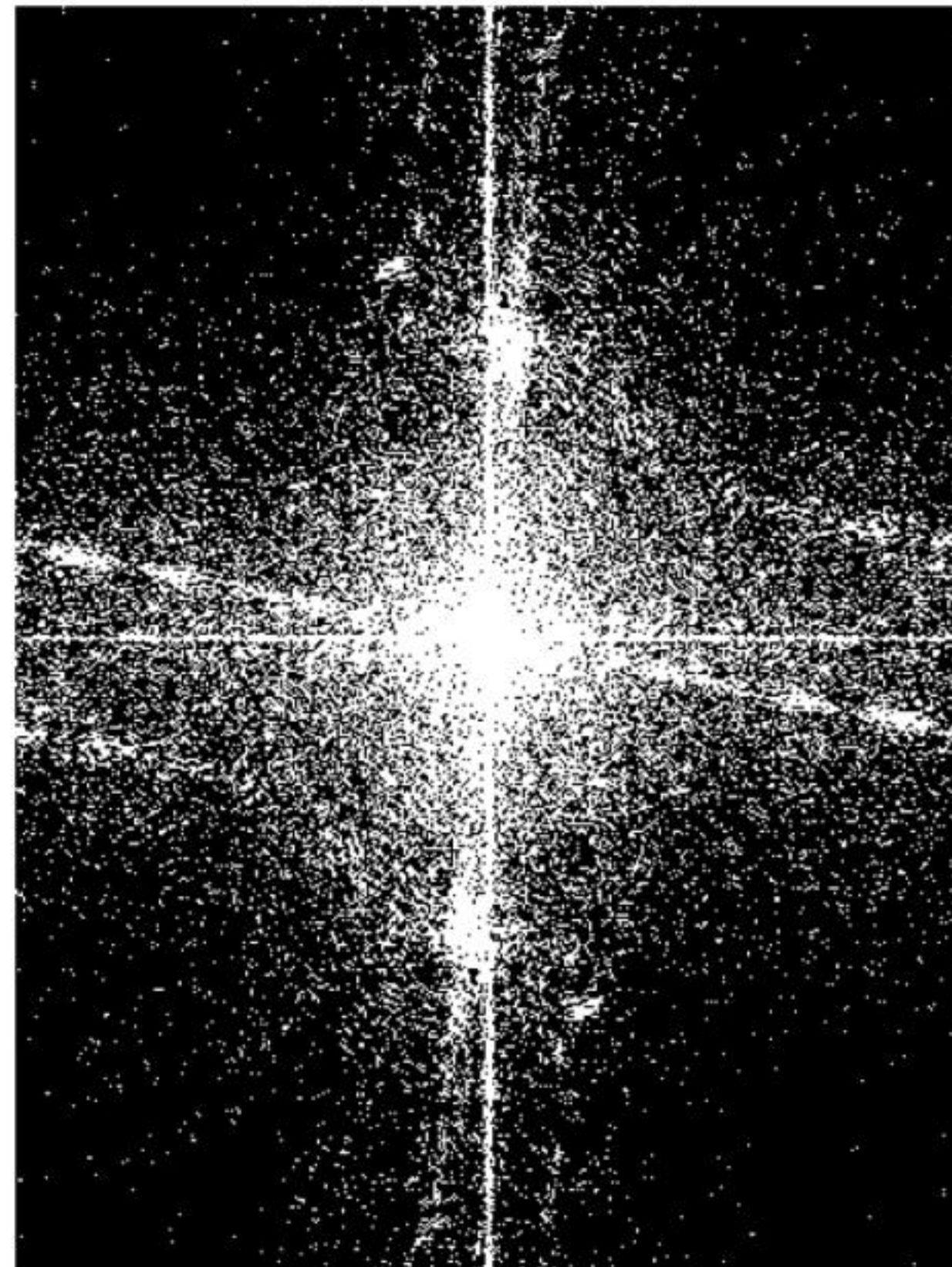
Kept coefficients, 1 %



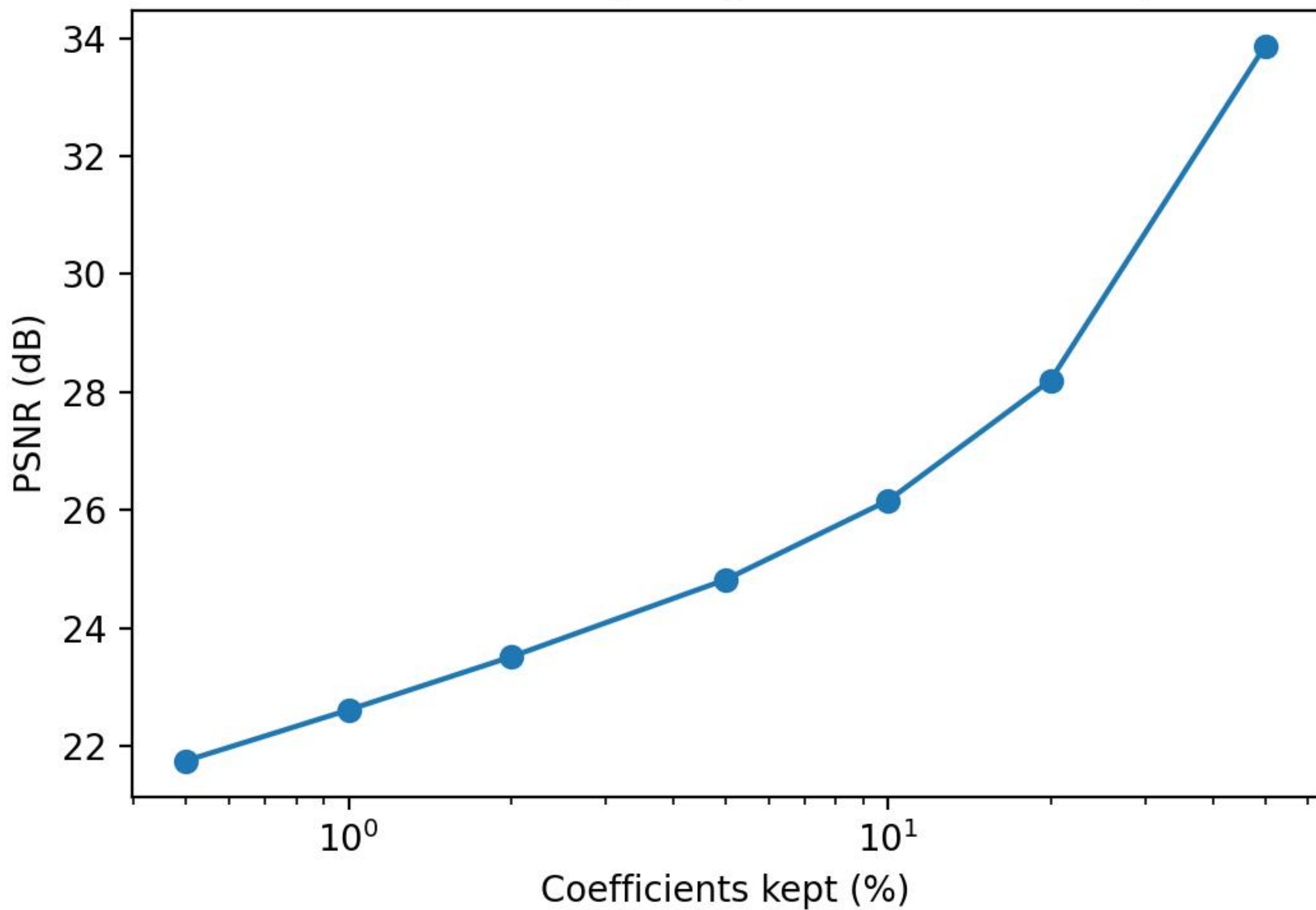
Kept coefficients, 5 %



Kept coefficients, 20 %



Reconstruction quality vs. coefficients kept



Mean face



PC 1



PC 2



PC 3



PC 4



PC 5



PC 6



PC 7



PC 8



PC 9



PC 10



PC 11



Original (p0, Happy)



k=1, MSE=0.0145



k=5, MSE=0.0107



k=10, MSE=0.0066



k=20, MSE=0.0035



k=50, MSE=0.0014



k=100, MSE=0.0006



k=148, MSE=0.0000



Original (p3, Neutral)



k=1, MSE=0.0264



k=5, MSE=0.0146



k=10, MSE=0.0085



k=20, MSE=0.0054



k=50, MSE=0.0026



k=100, MSE=0.0001



k=148, MSE=0.0000



Original (p9, Surprised)



k=1, MSE=0.0186



k=5, MSE=0.0095



k=10, MSE=0.0058



k=20, MSE=0.0050



k=50, MSE=0.0021



k=100, MSE=0.0003



k=148, MSE=0.0000



Original (p18, Anger)



k=1, MSE=0.0569



k=5, MSE=0.0065



k=10, MSE=0.0061



k=20, MSE=0.0044



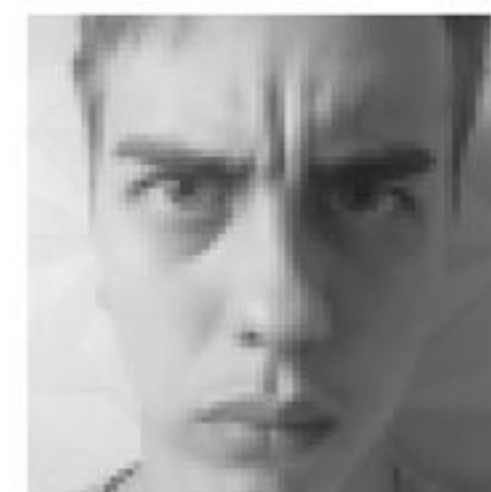
k=50, MSE=0.0017



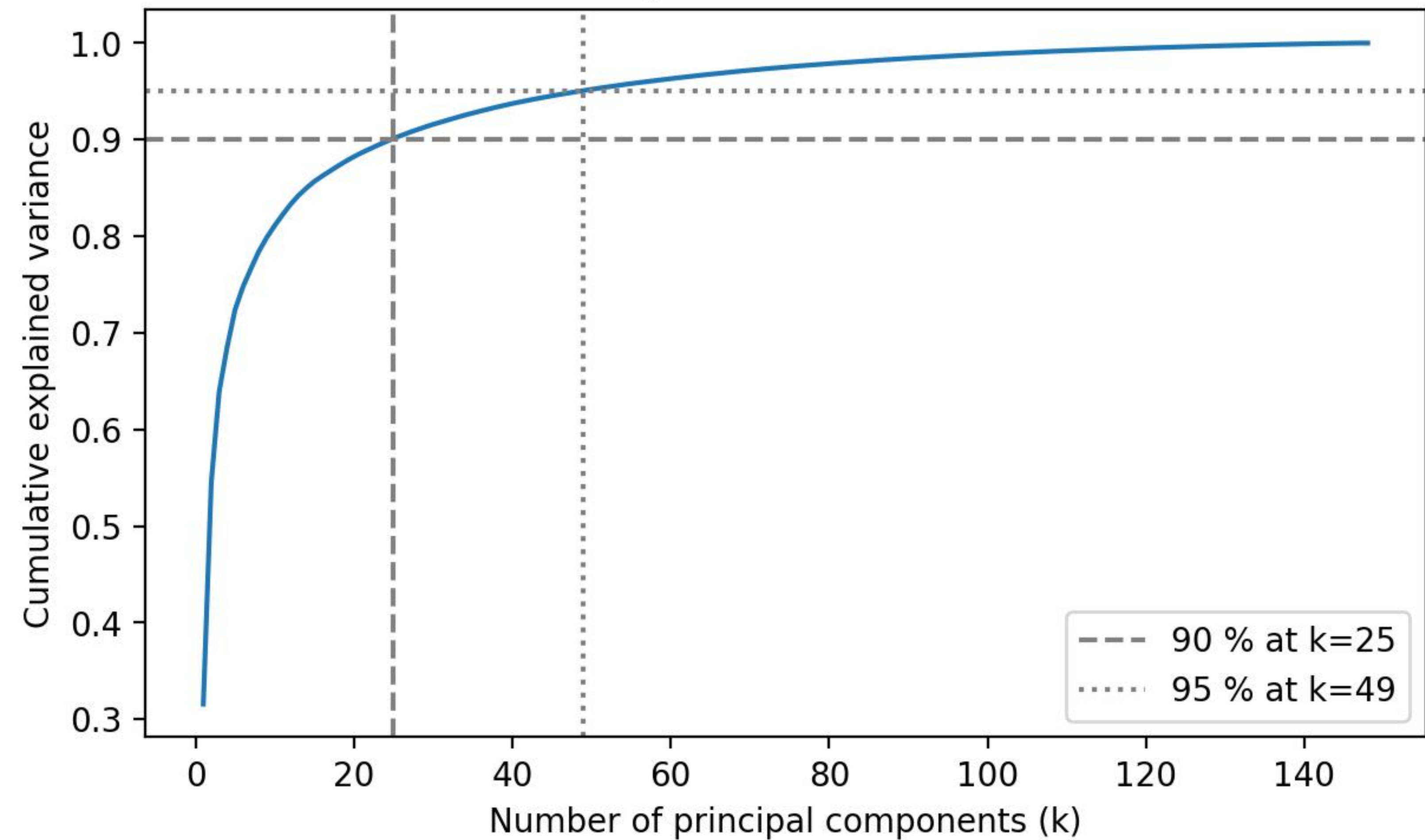
k=100, MSE=0.0005



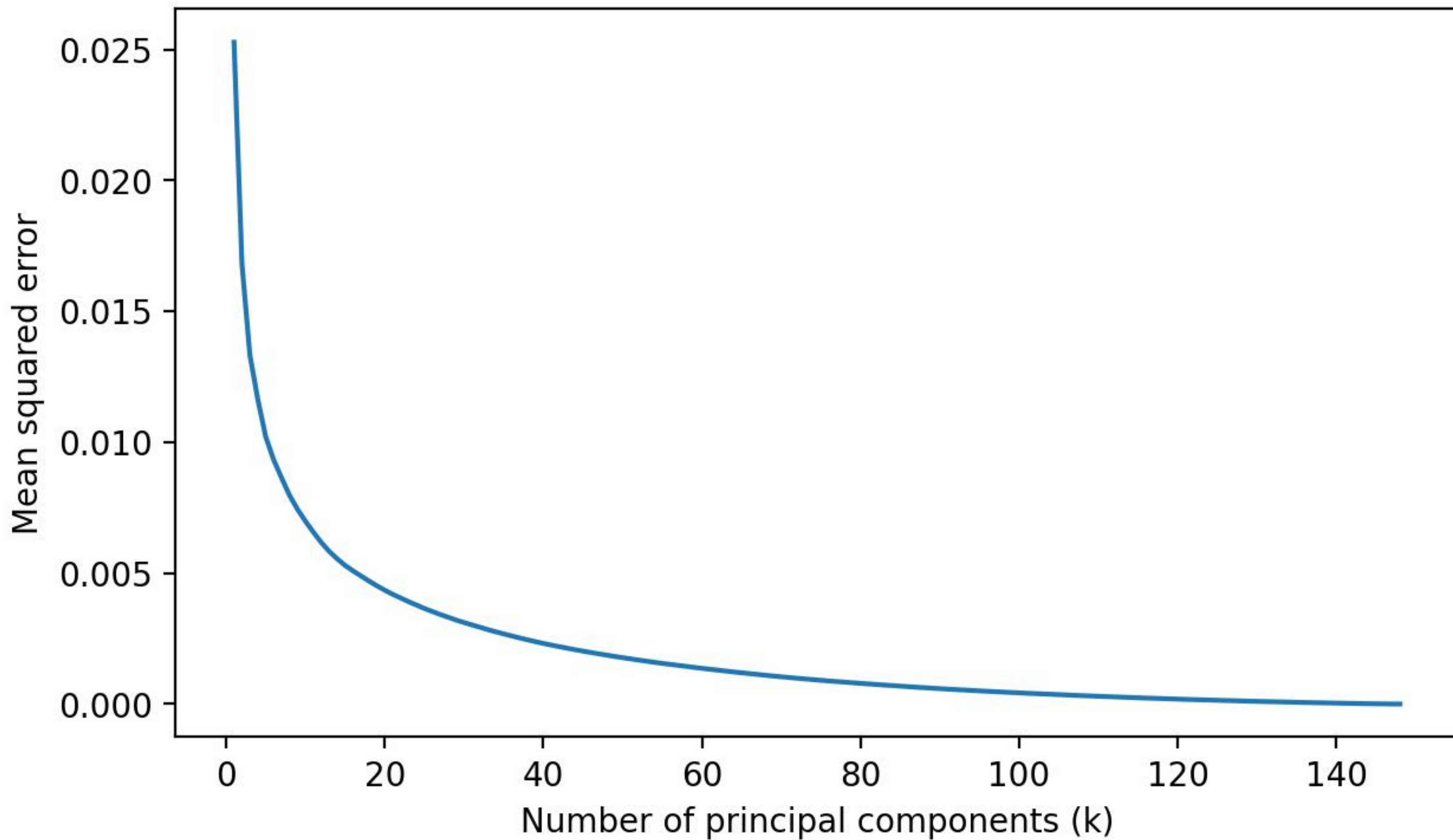
k=148, MSE=0.0000



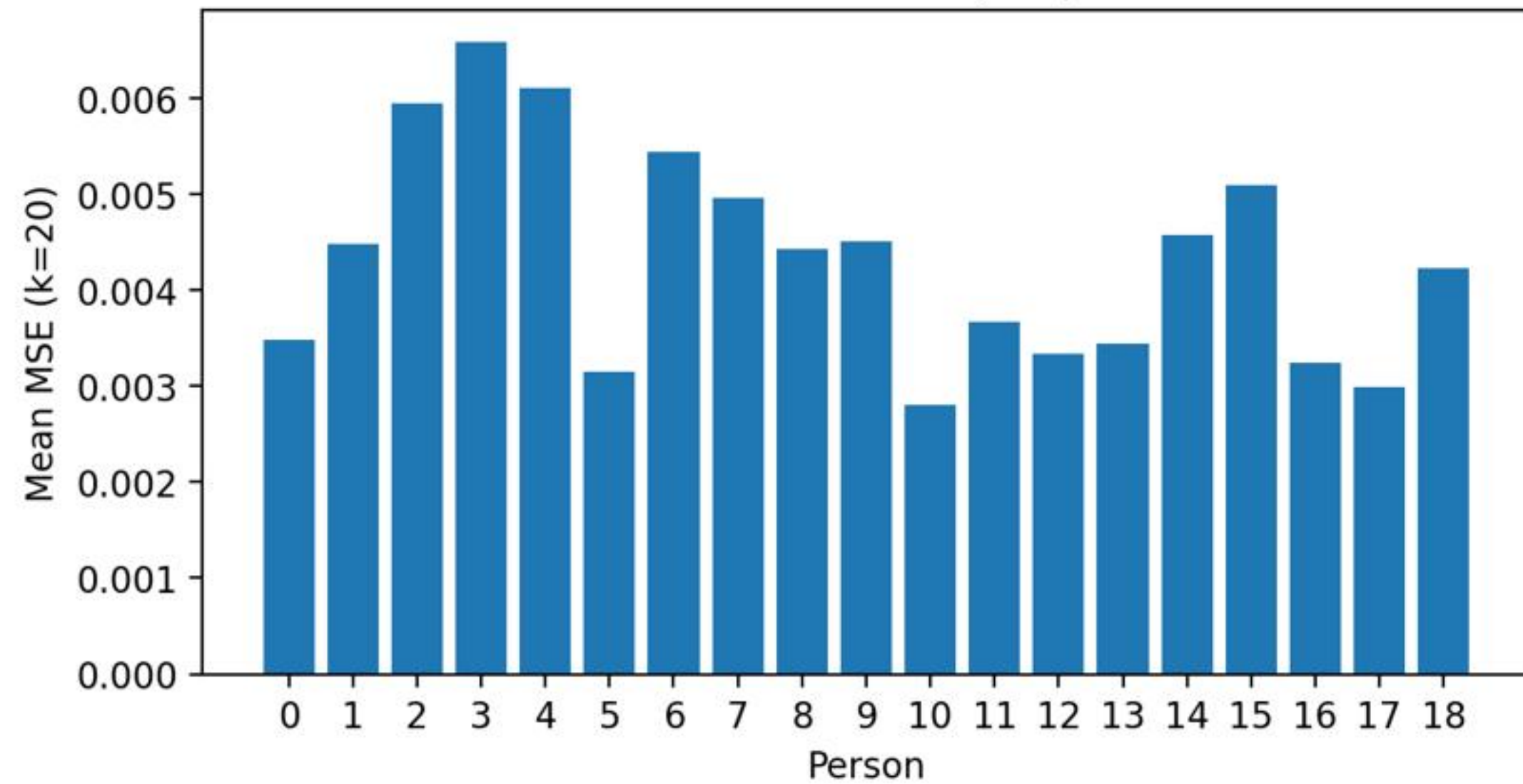
Explained variance



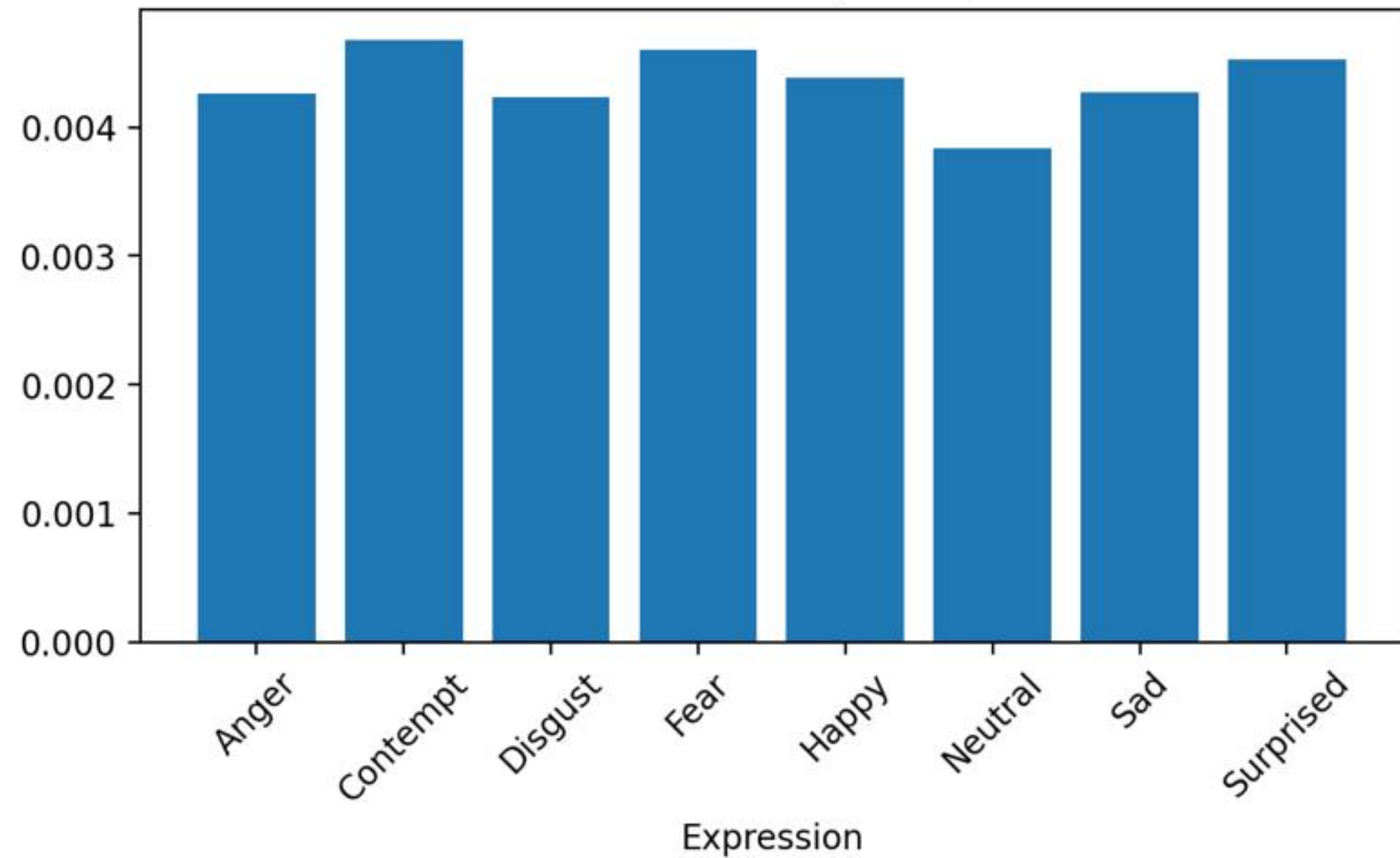
Reconstruction error vs. k



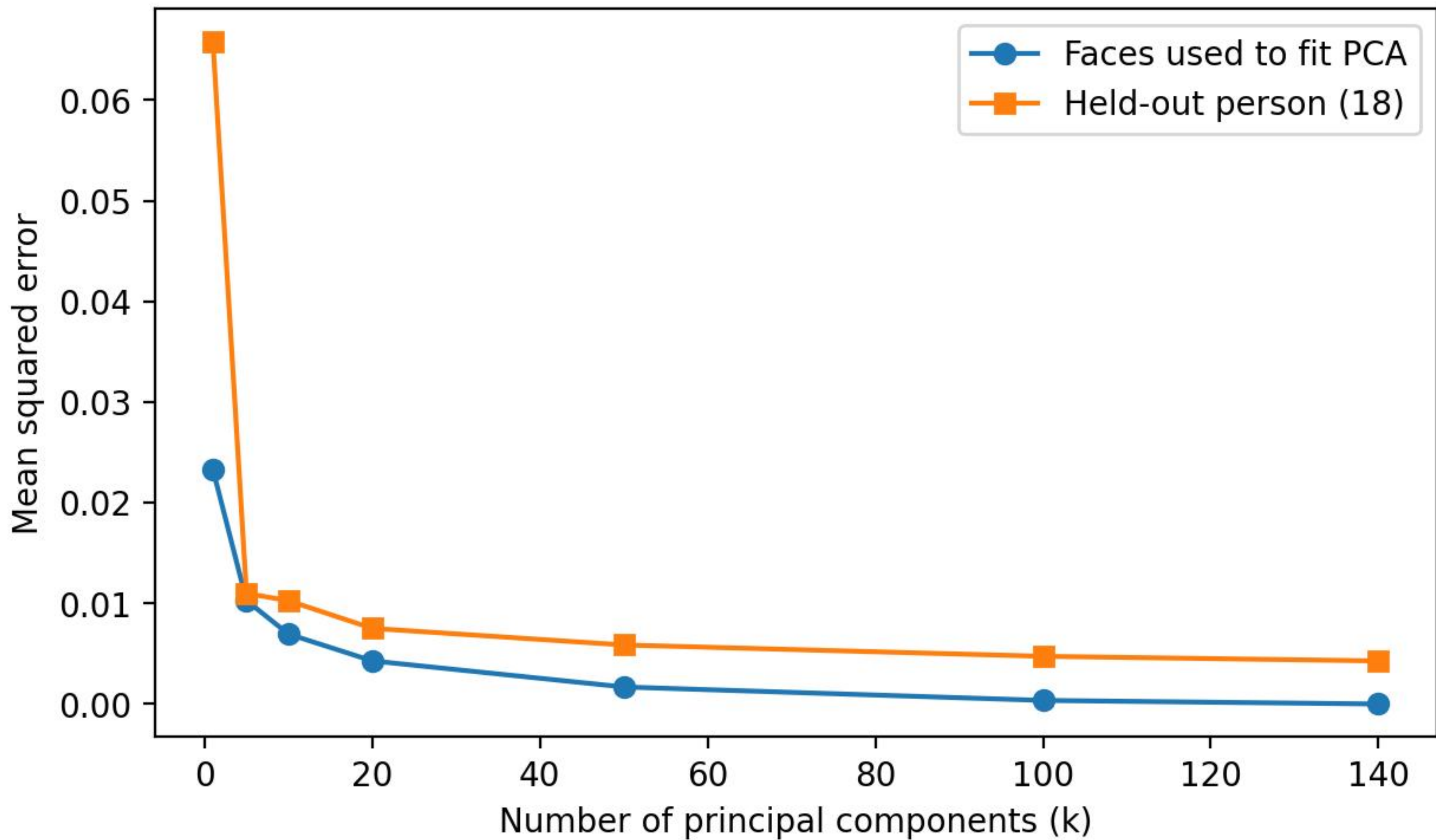
Reconstruction error per person



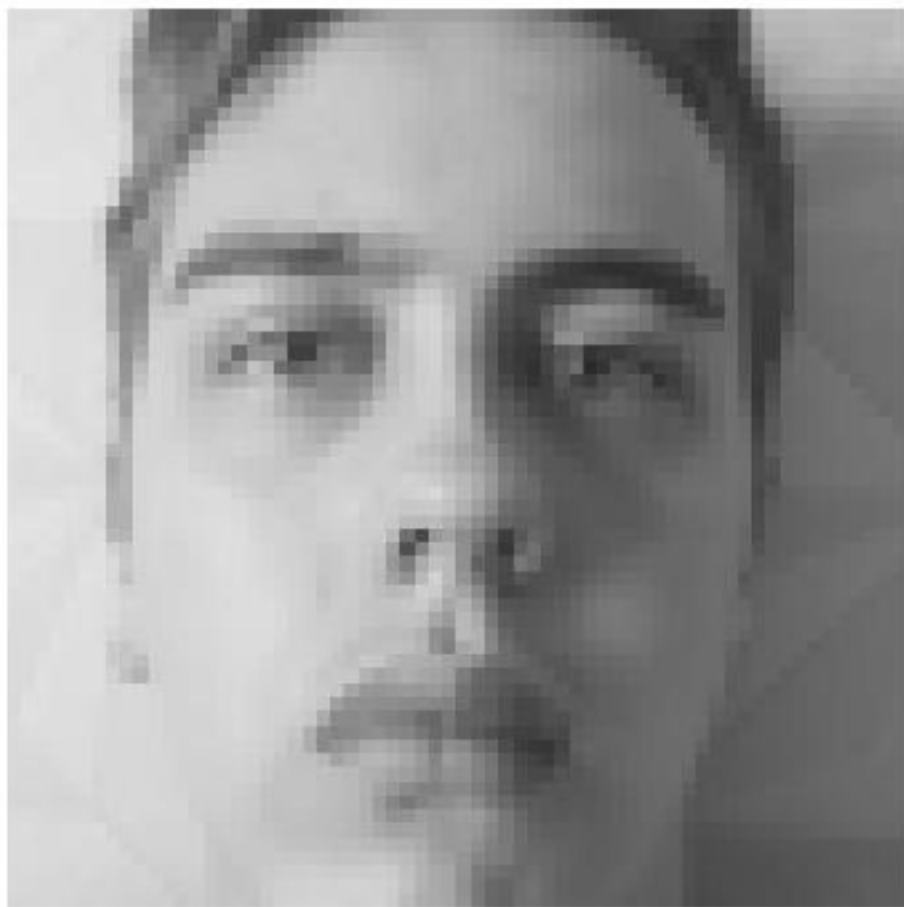
Reconstruction error per expression



Known vs. unseen faces



Original (held out)



$k=20$, $MSE=0.0067$



$k=50$, $MSE=0.0054$



$k=140$, $MSE=0.0037$



Simple (person 18)



Gradient magnitude (Sobel)



HOG, 106596 features



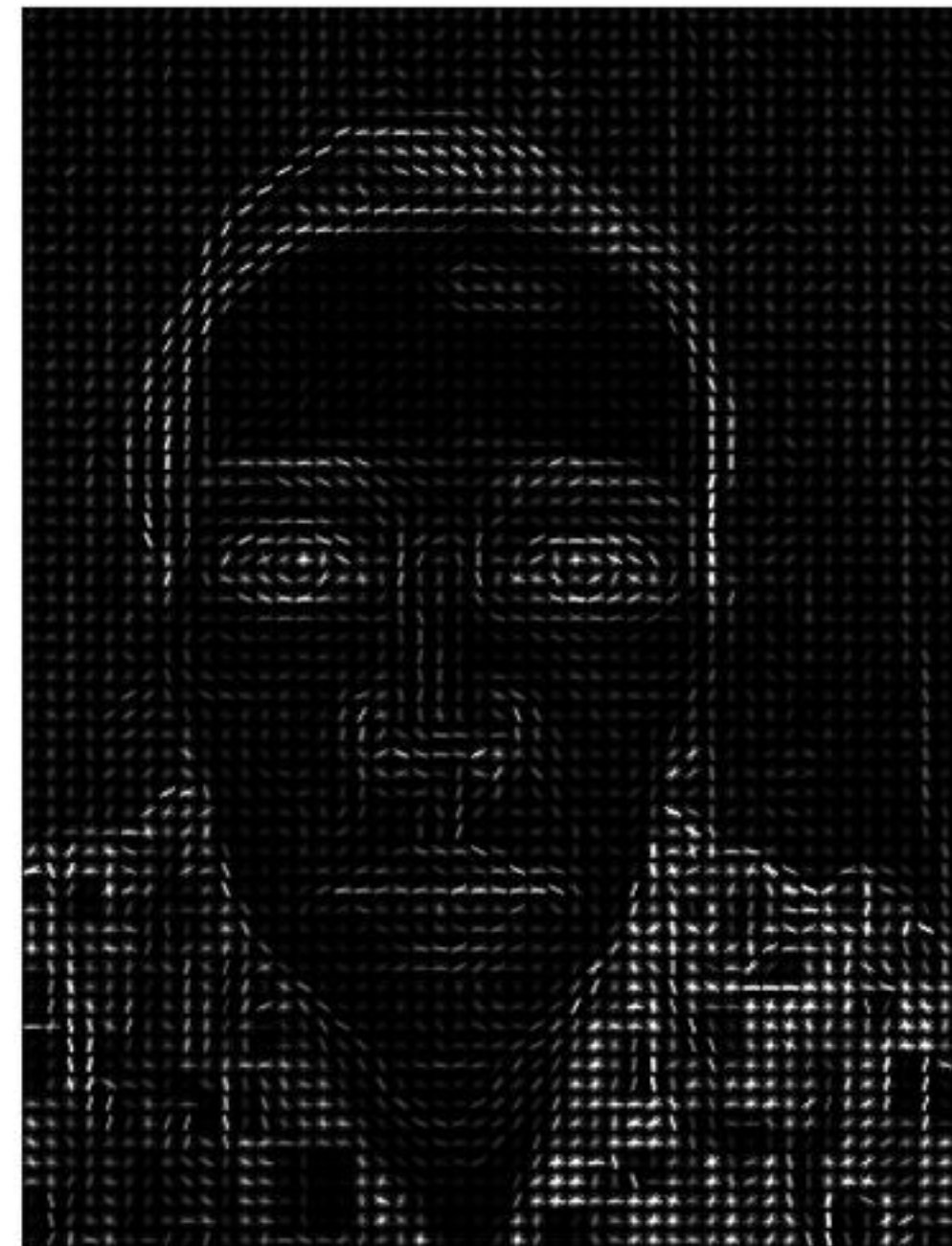
Medium (person 4)



Gradient magnitude (Sobel)



HOG, 106596 features



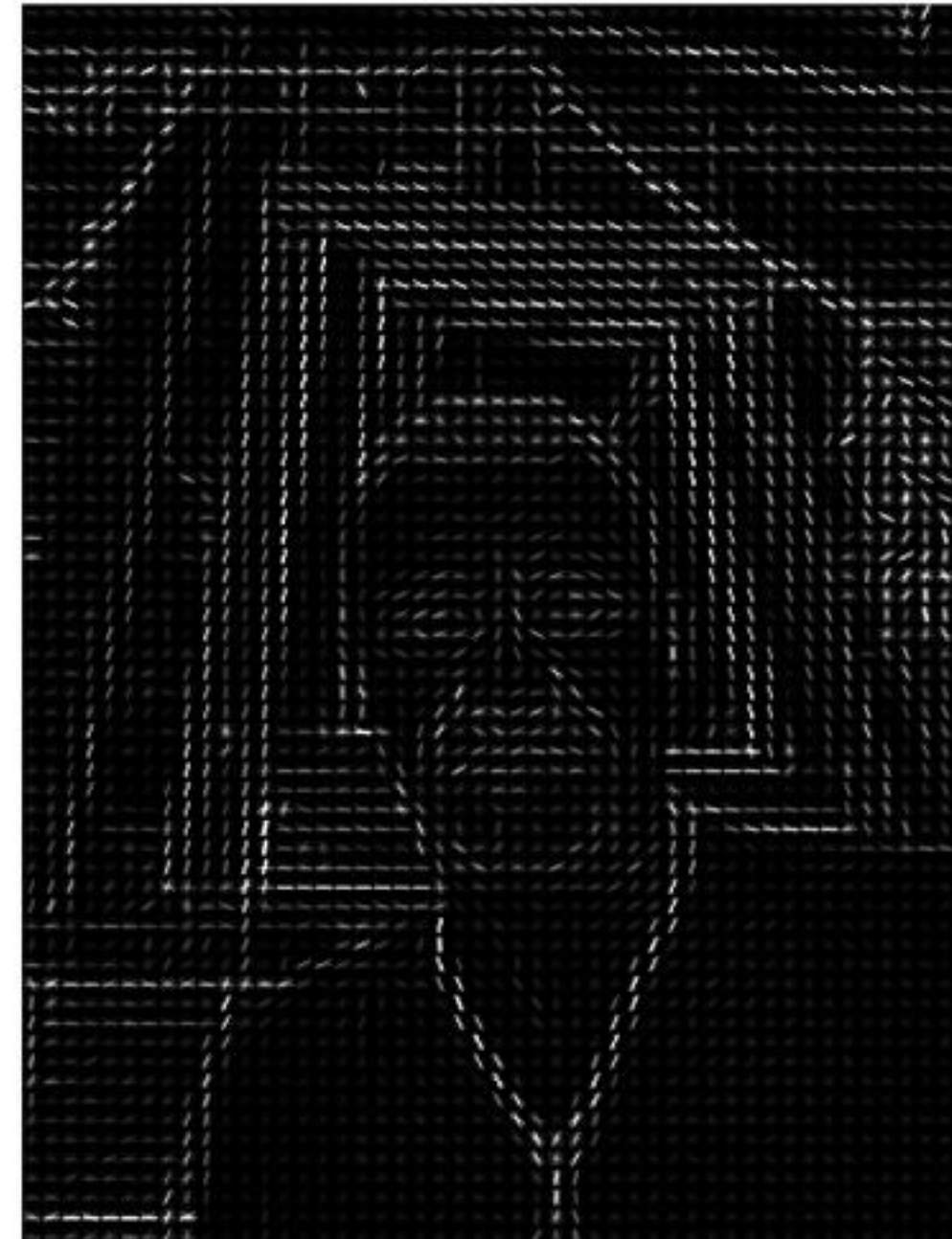
Complex (person 5)



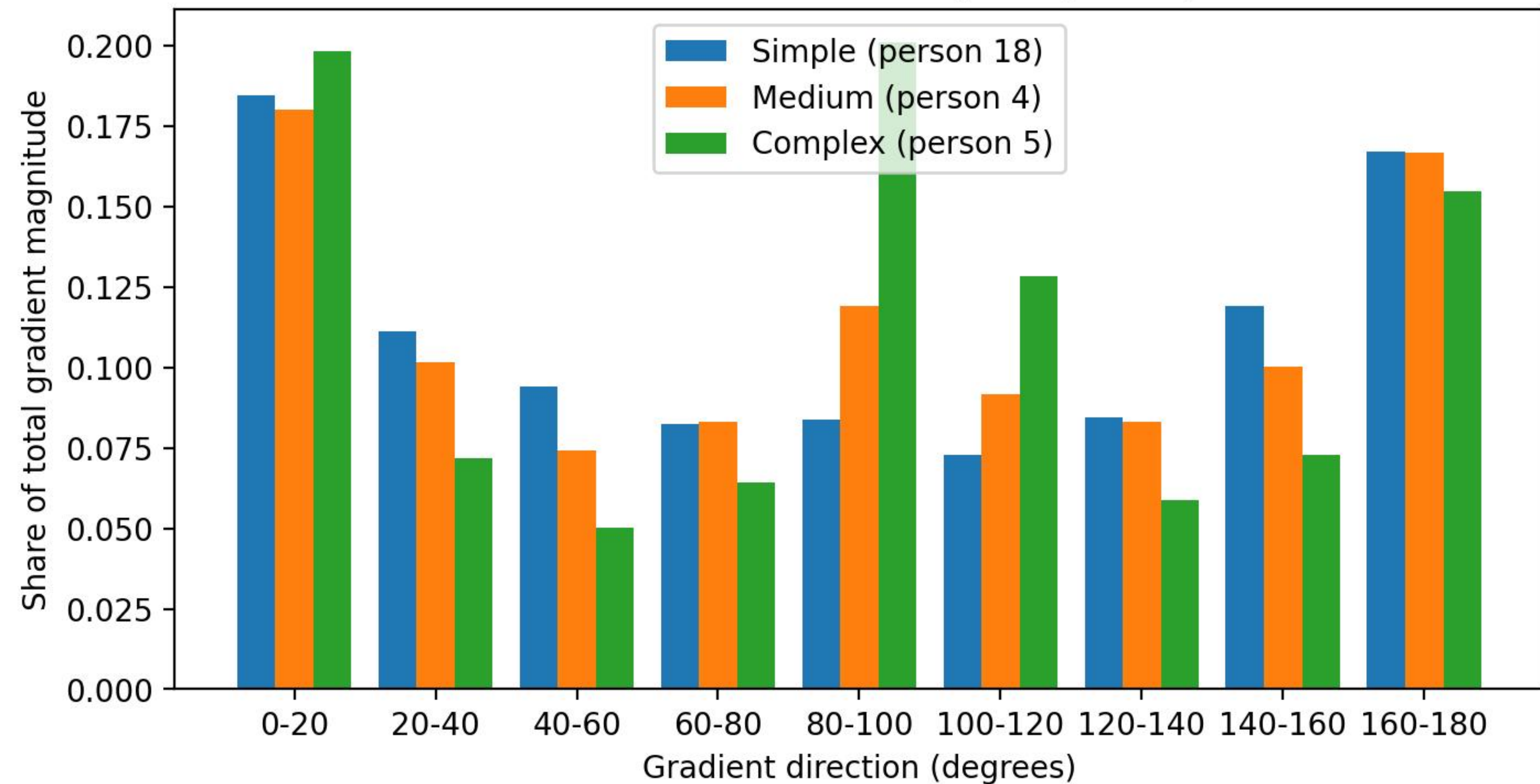
Gradient magnitude (Sobel)



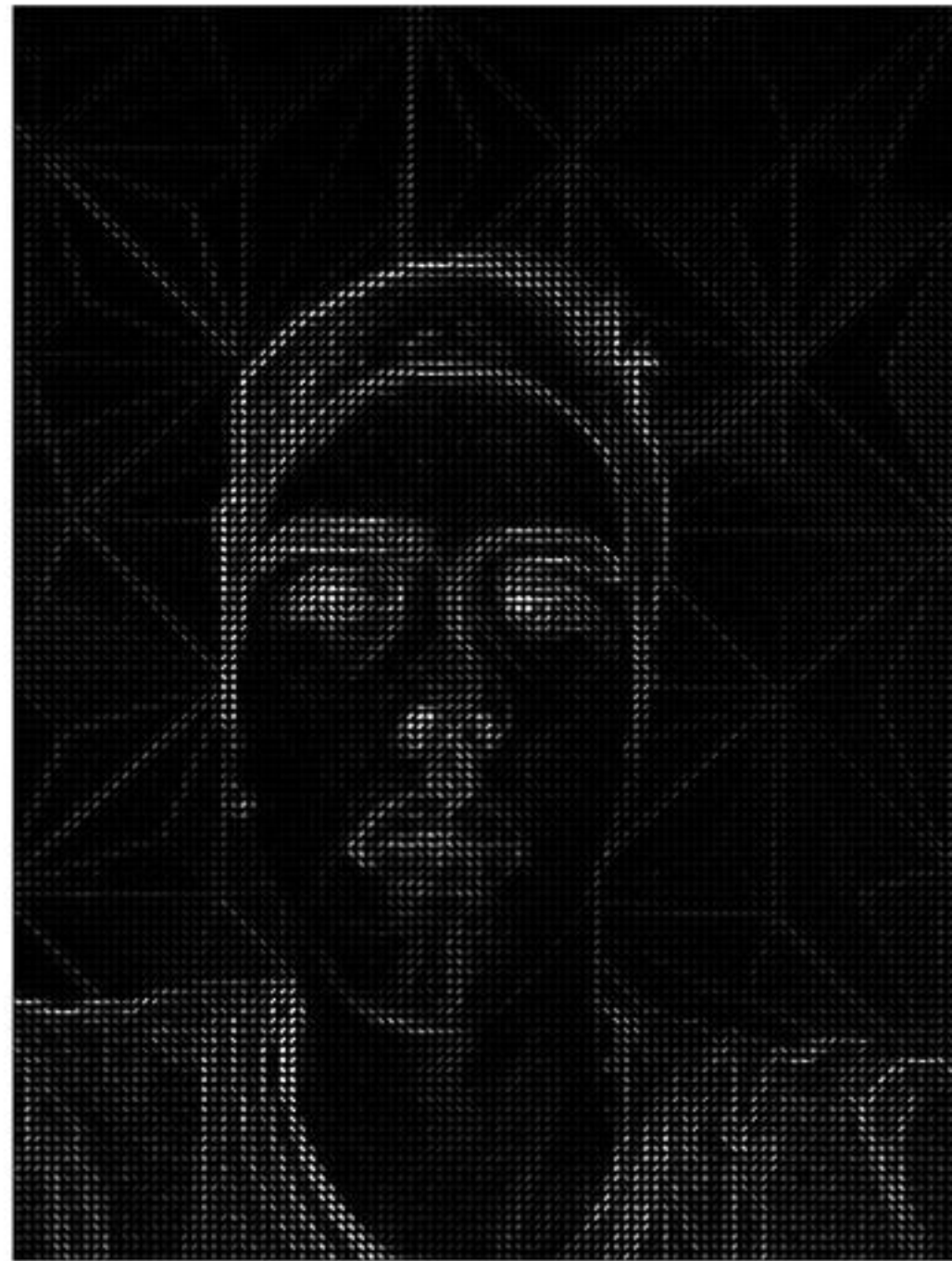
HOG, 106596 features



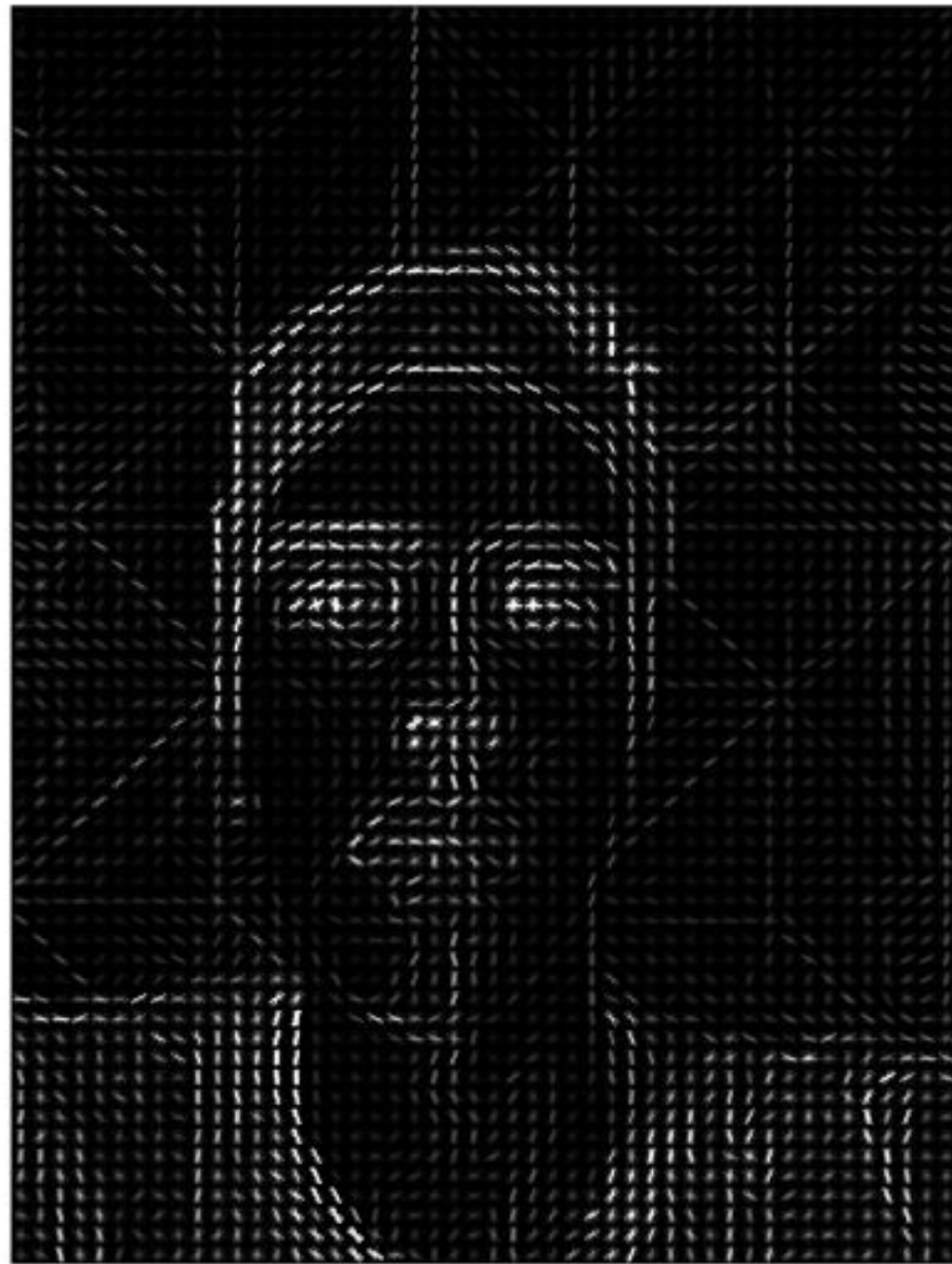
Global orientation histogram (9 bins)



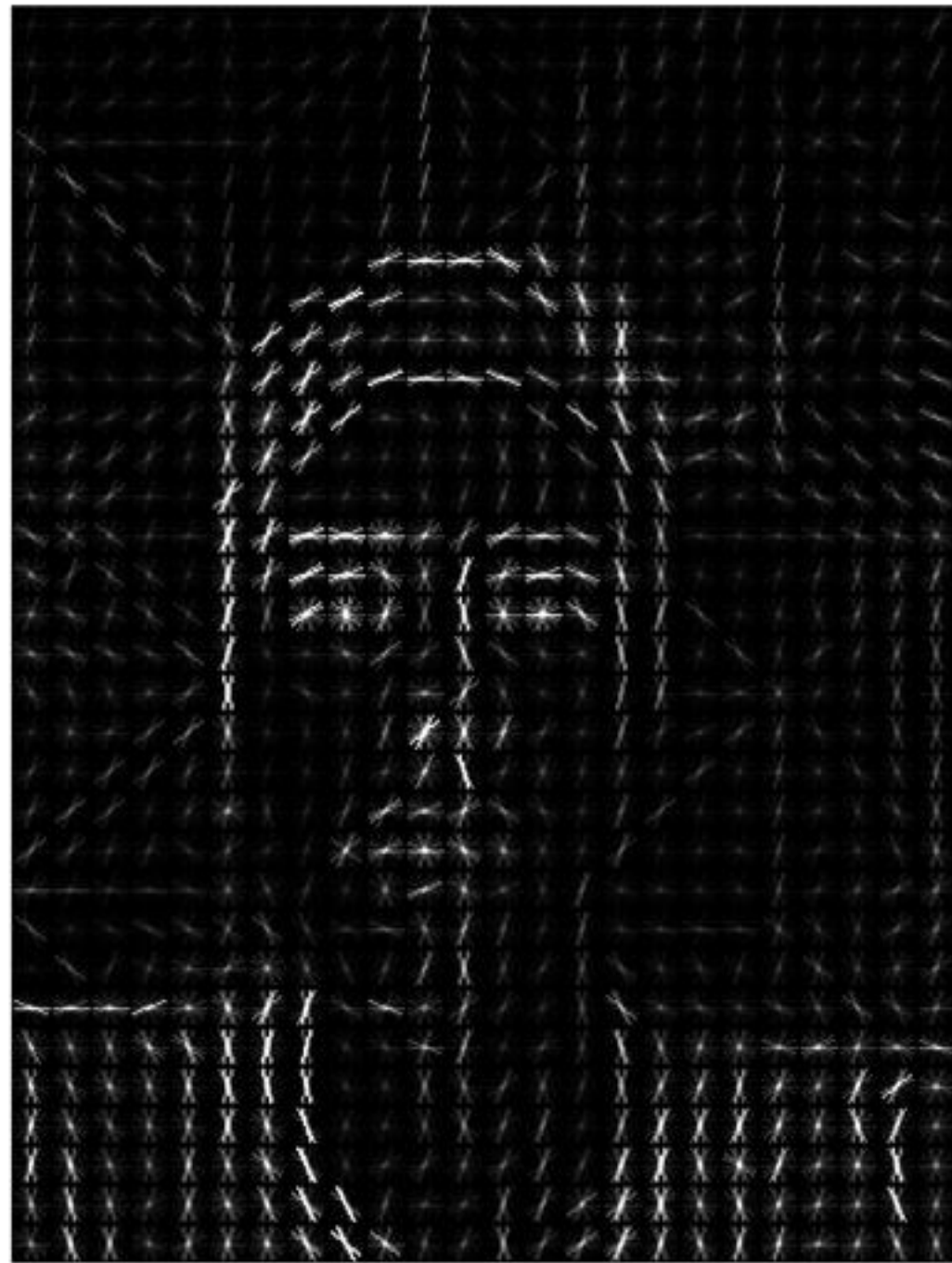
Cell 4x4, 434340 features



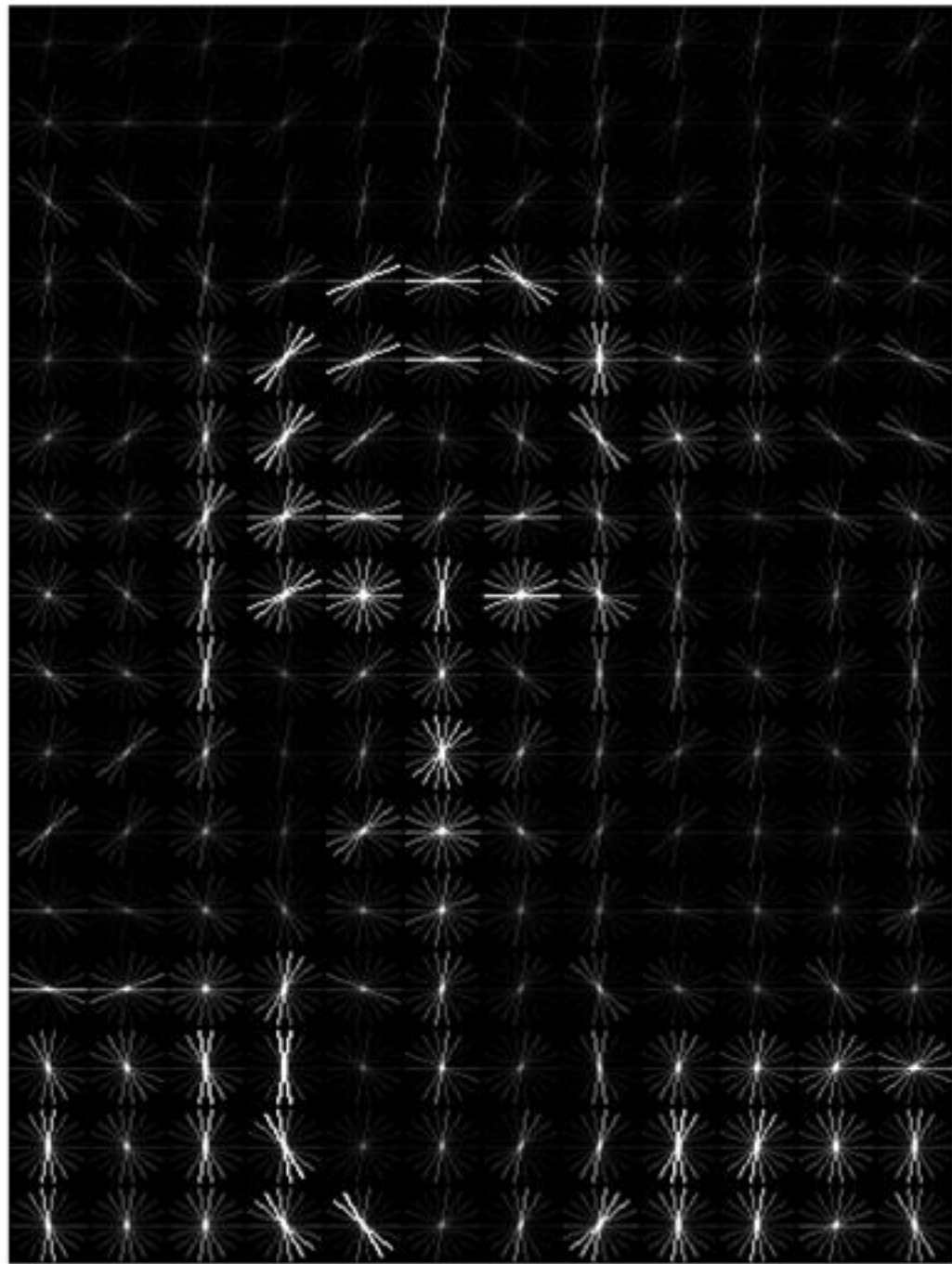
Cell 8x8, 106596 features



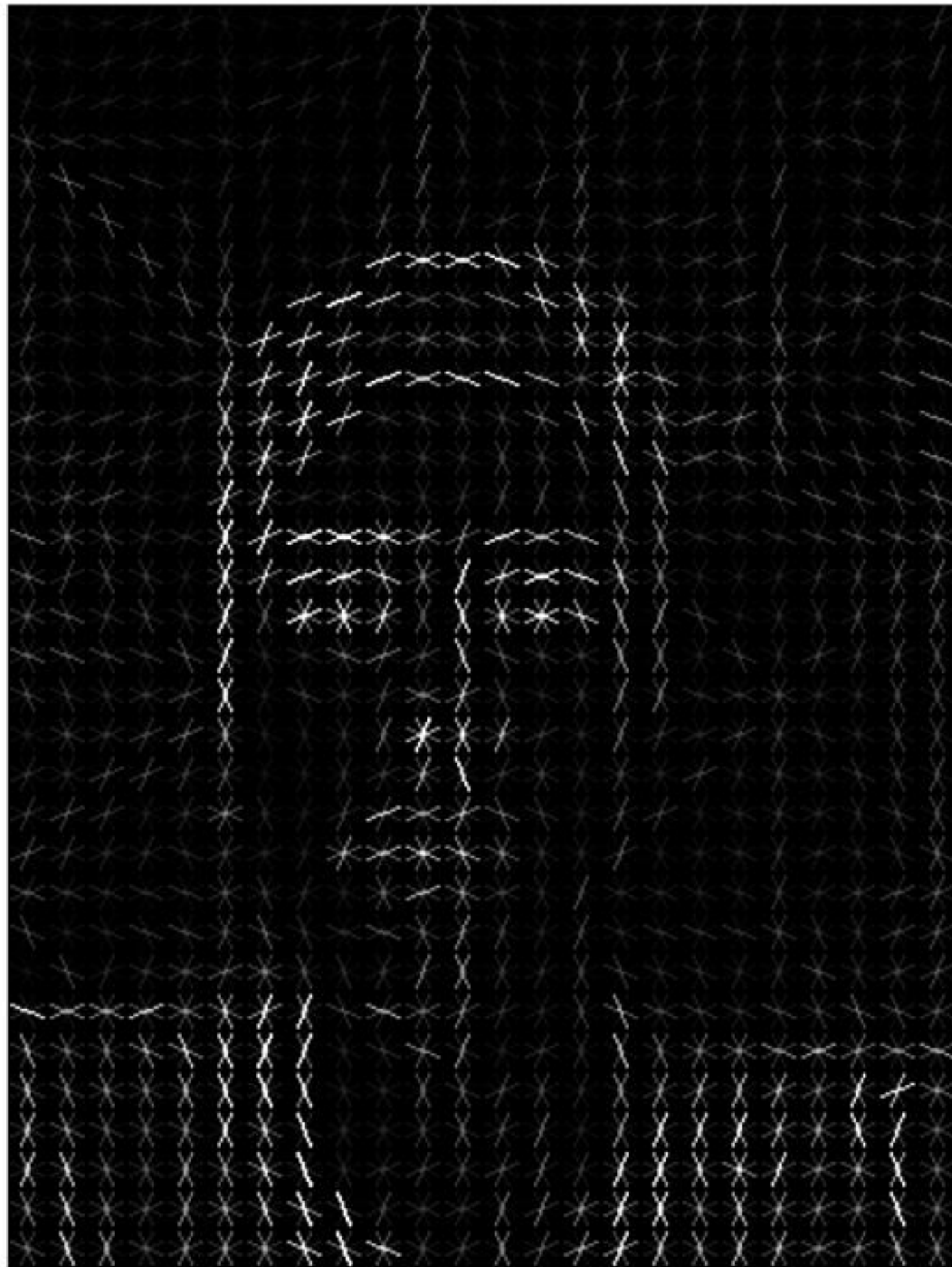
Cell 16x16, 25668 features



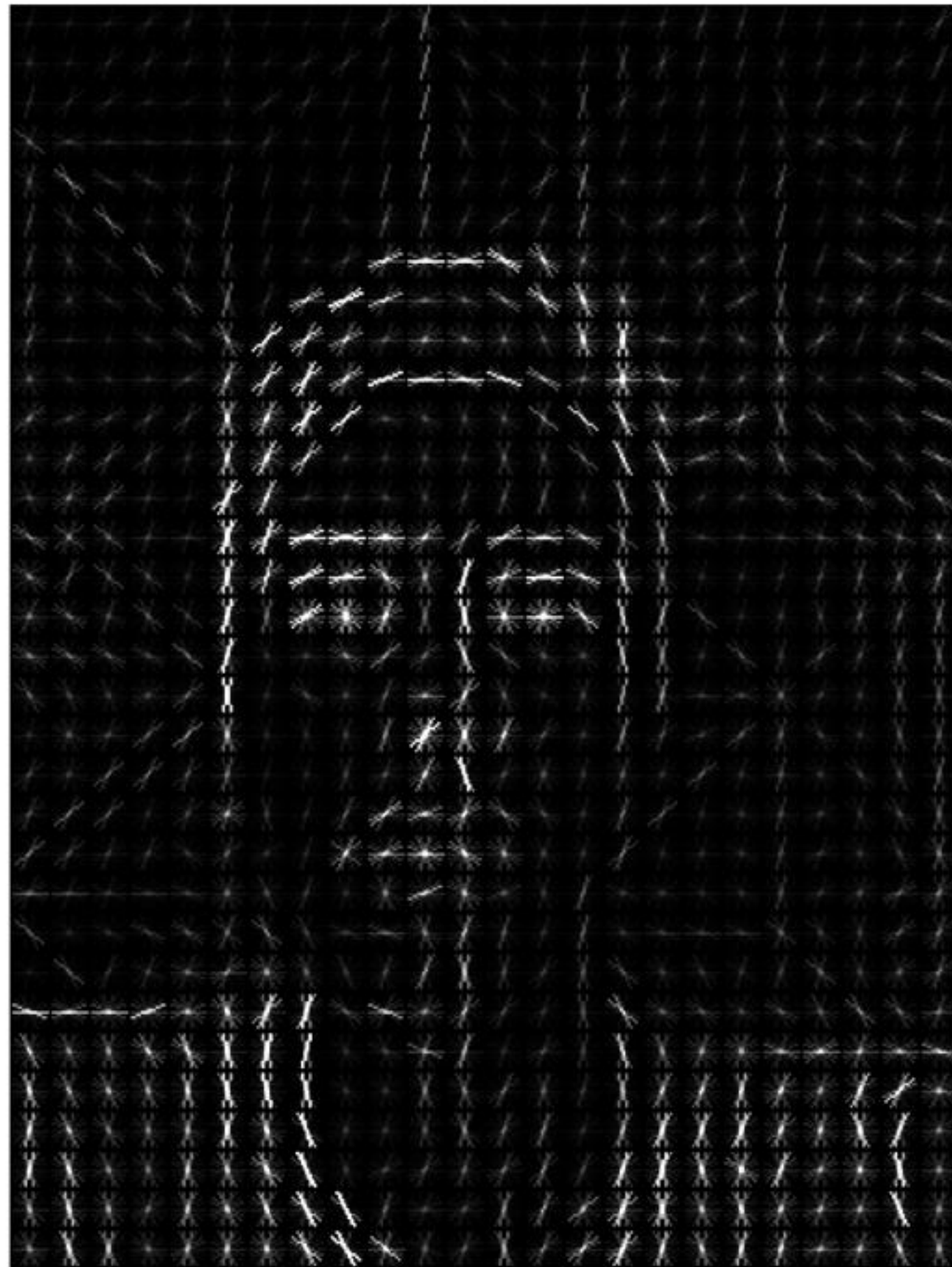
Cell 32x32, 5940 features



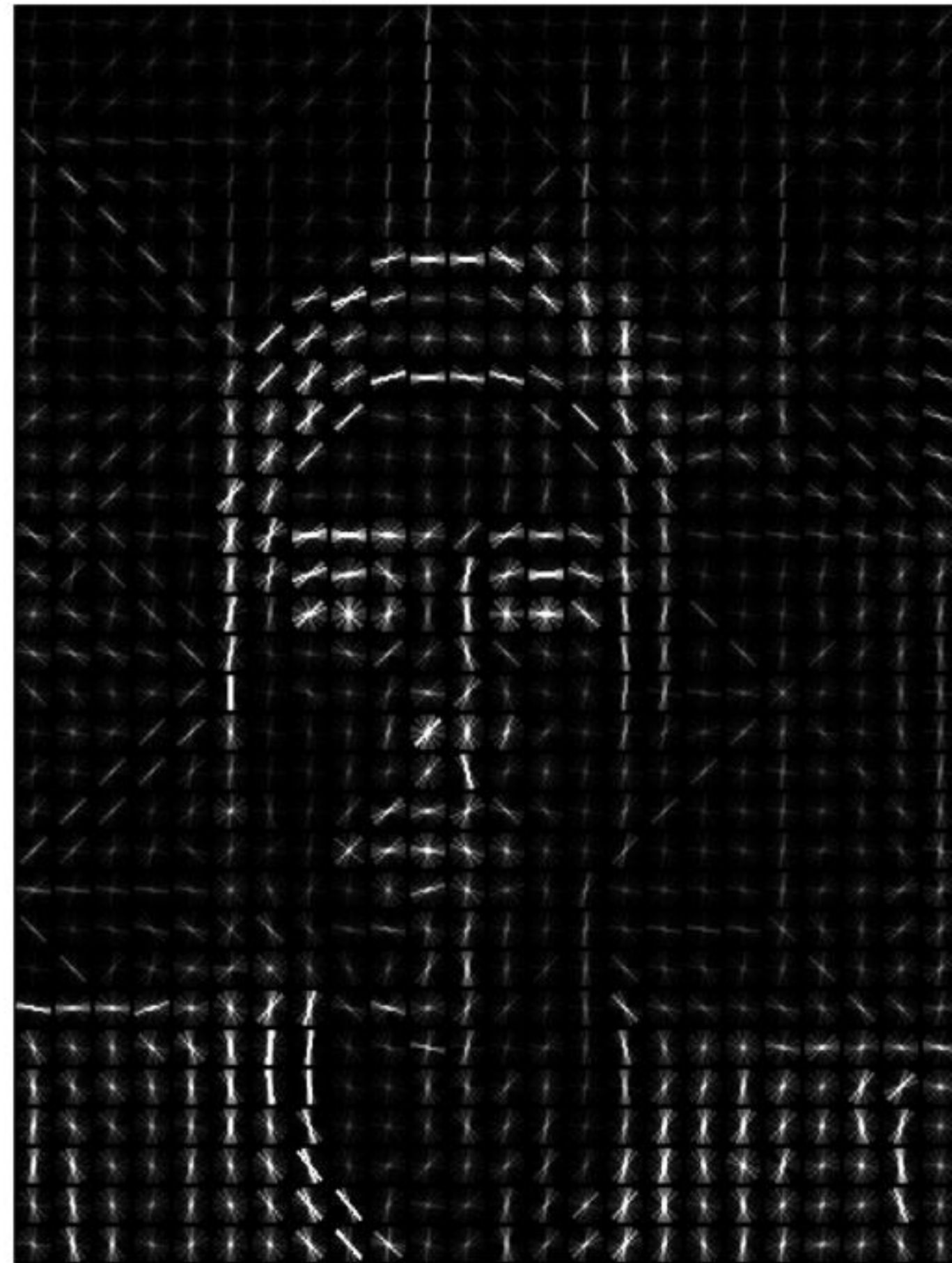
4 bins (cell 16x16), 11408 features



9 bins (cell 16x16), 25668 features



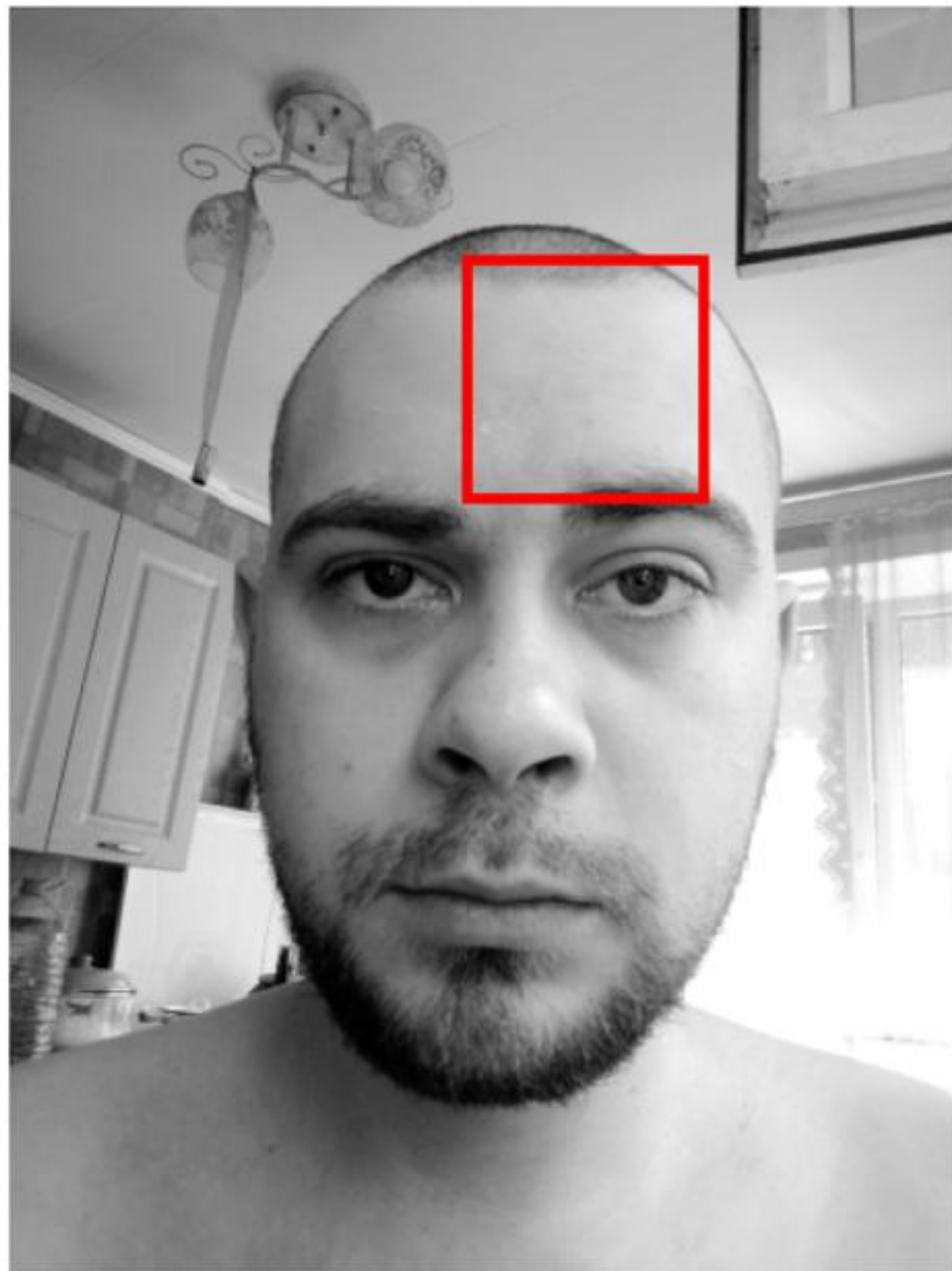
18 bins (cell 16x16), 51336 features



B: Hair (person 9)



C: Skin (person 3)



D: Fabric (person 4)



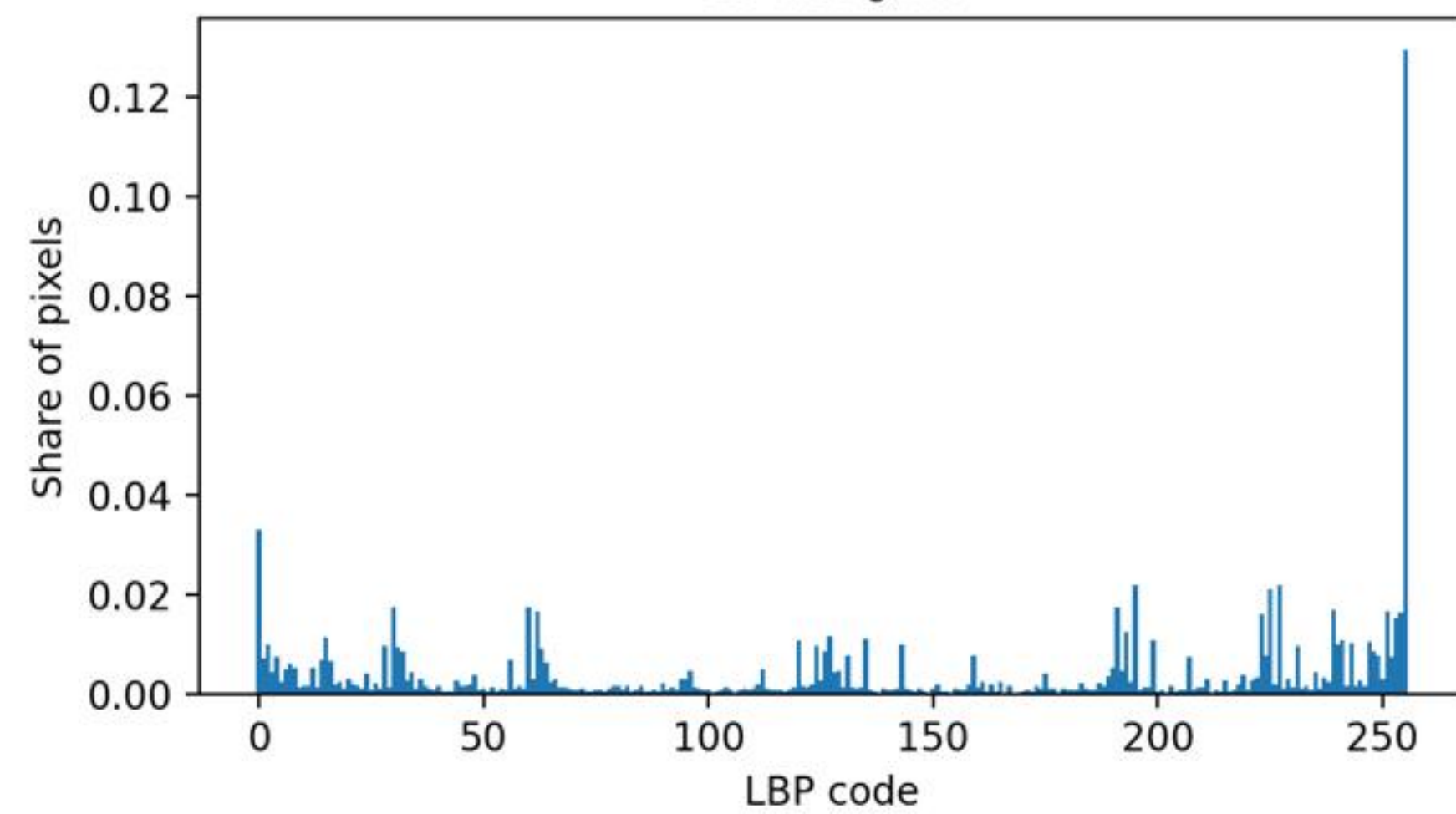
A: Face (person 18)



LBP image



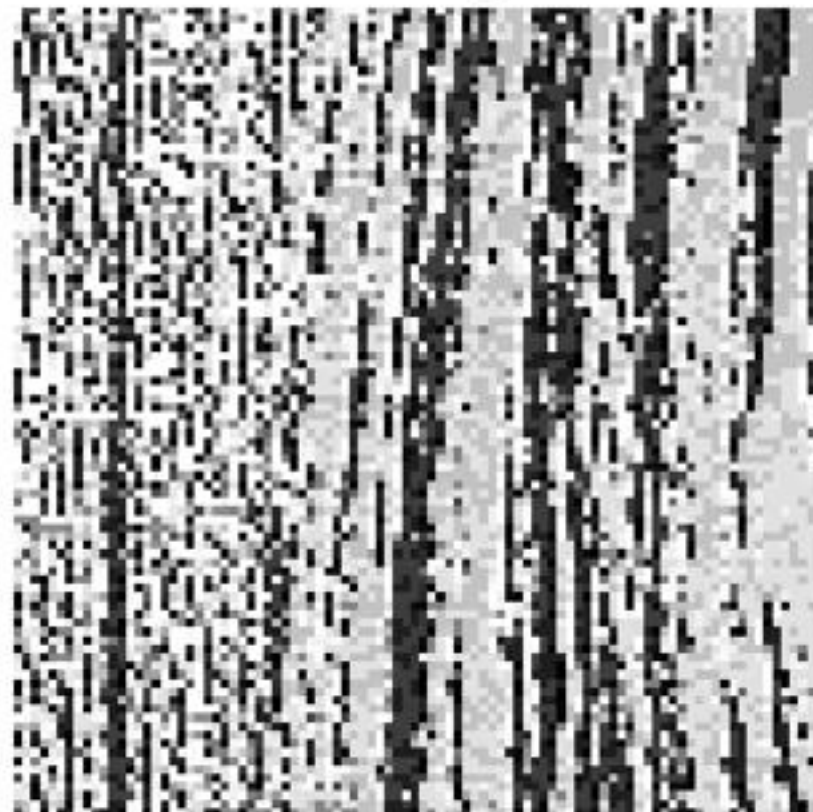
LBP histogram



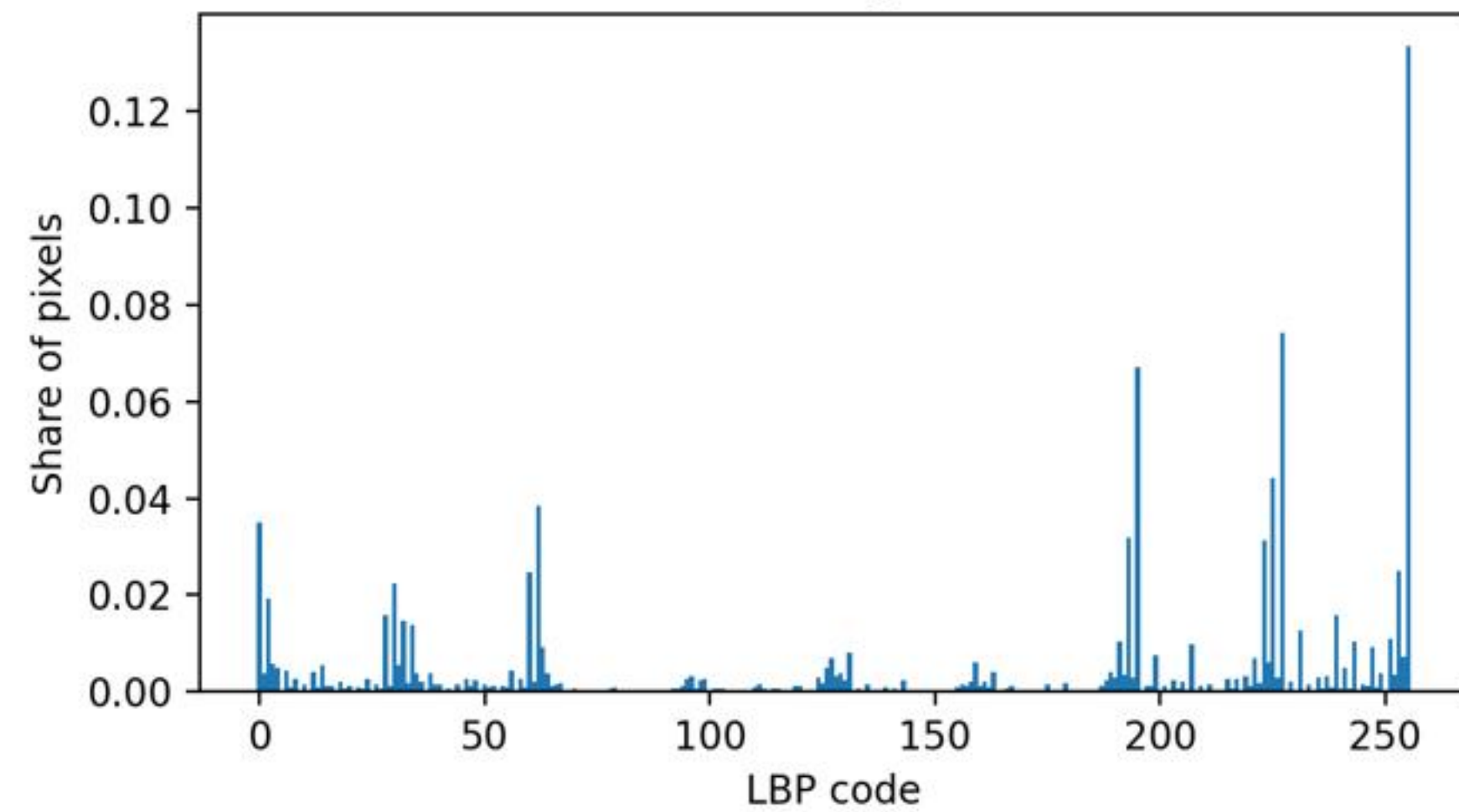
B: Hair (person 9)



LBP image



LBP histogram



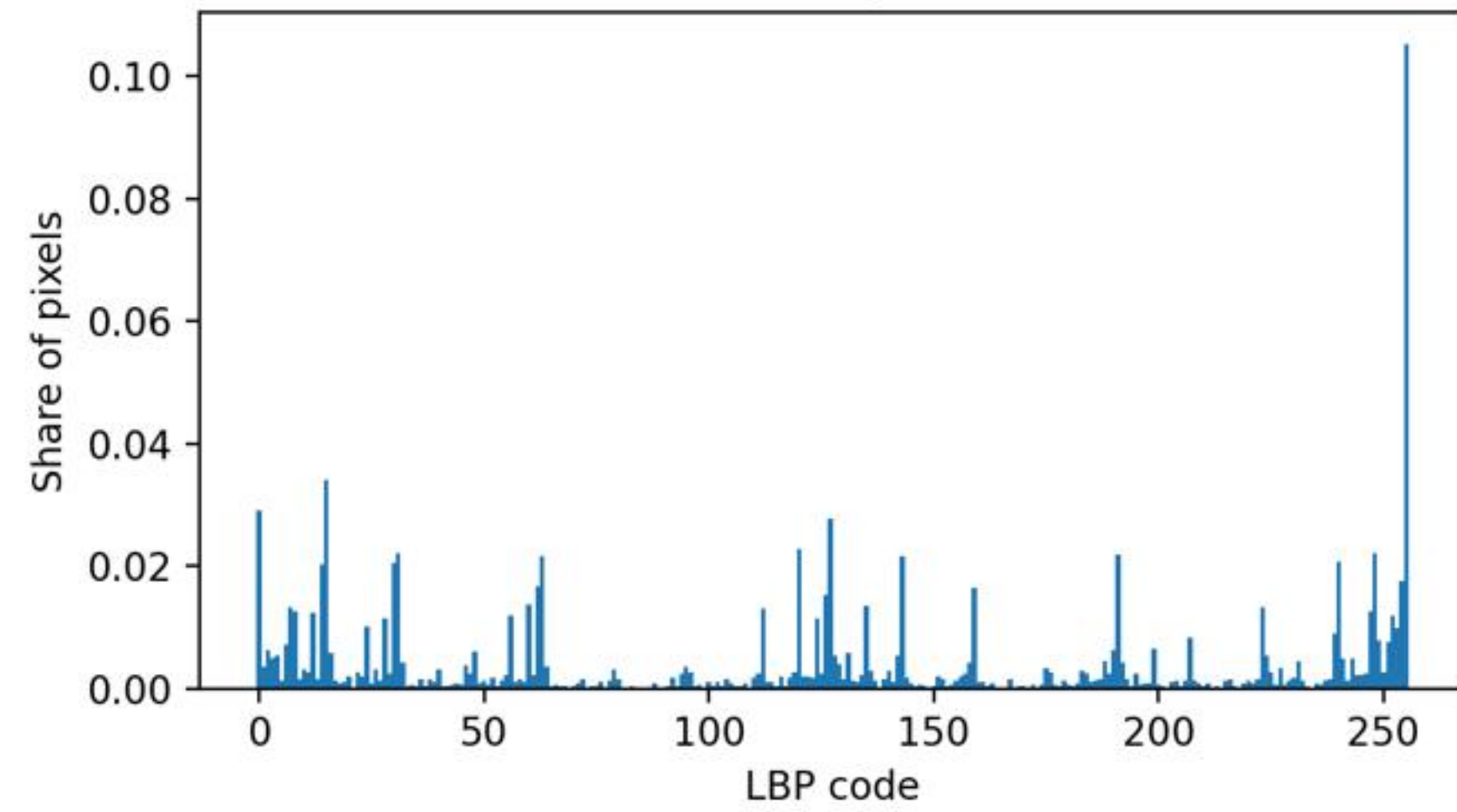
C: Skin (person 3)



LBP image



LBP histogram



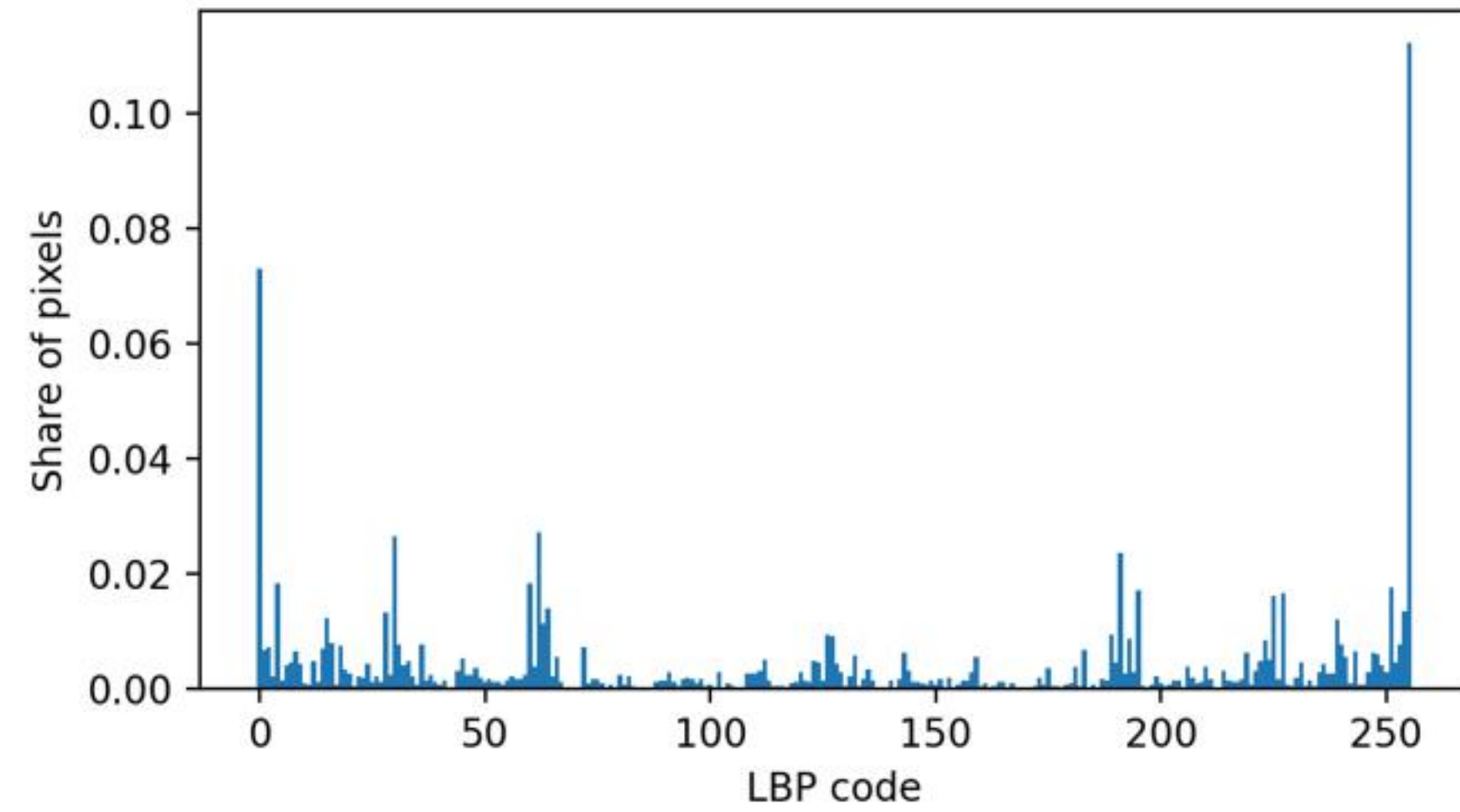
D: Fabric (person 4)



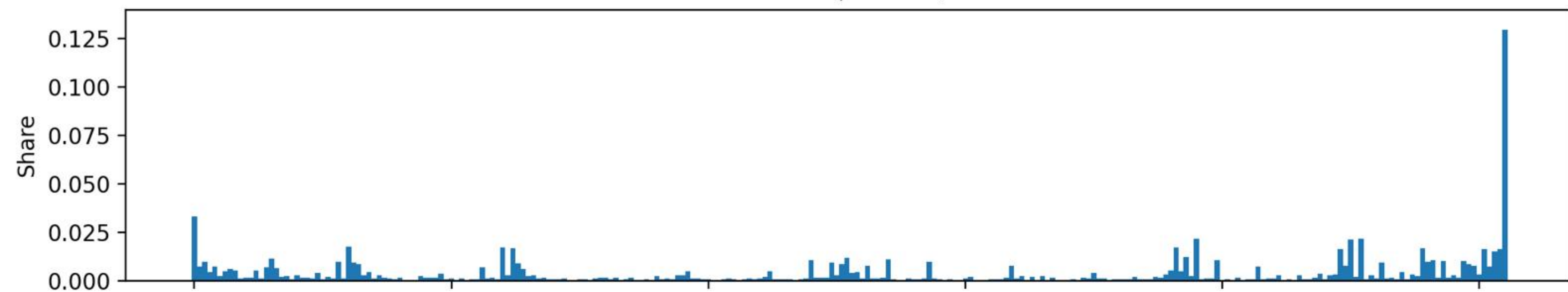
LBP image



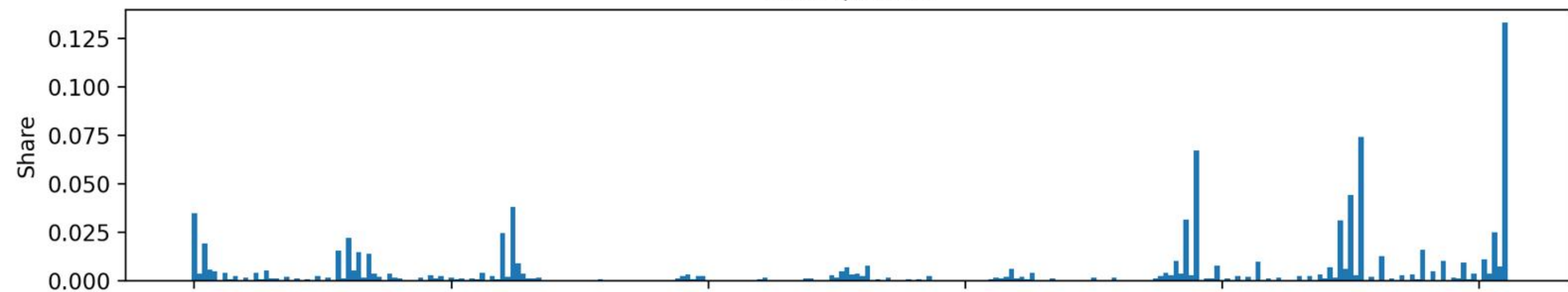
LBP histogram



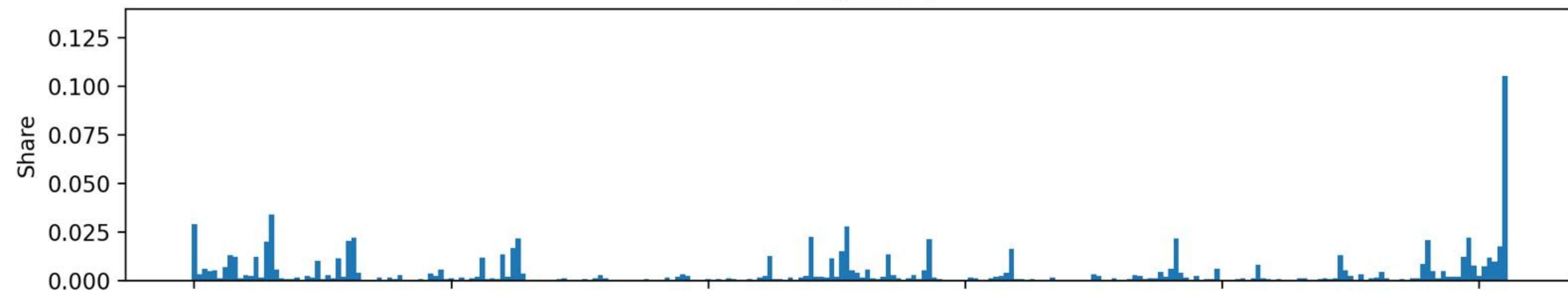
A: Face (person 18)



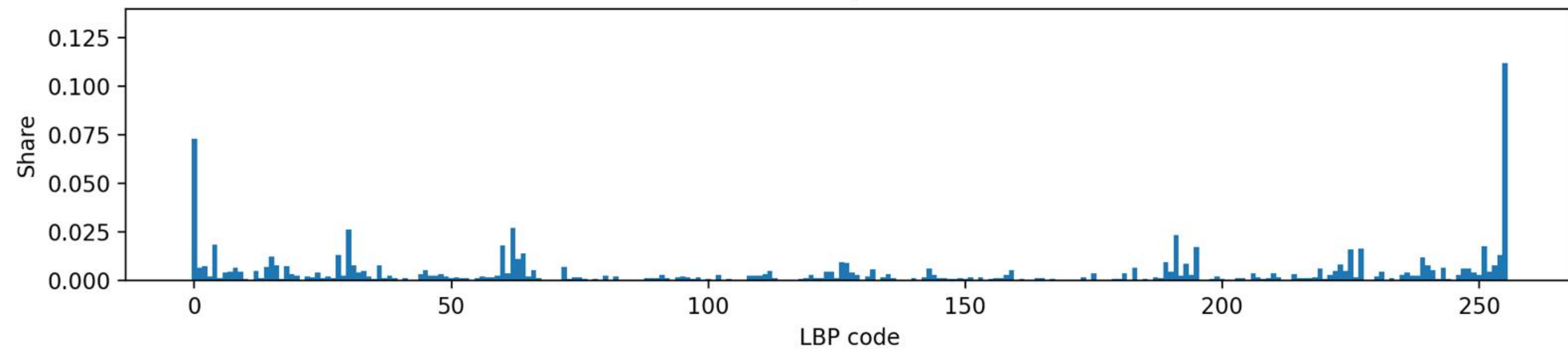
B: Hair (person 9)



C: Skin (person 3)



D: Fabric (person 4)



Person 3, Neutral: 38 blobs



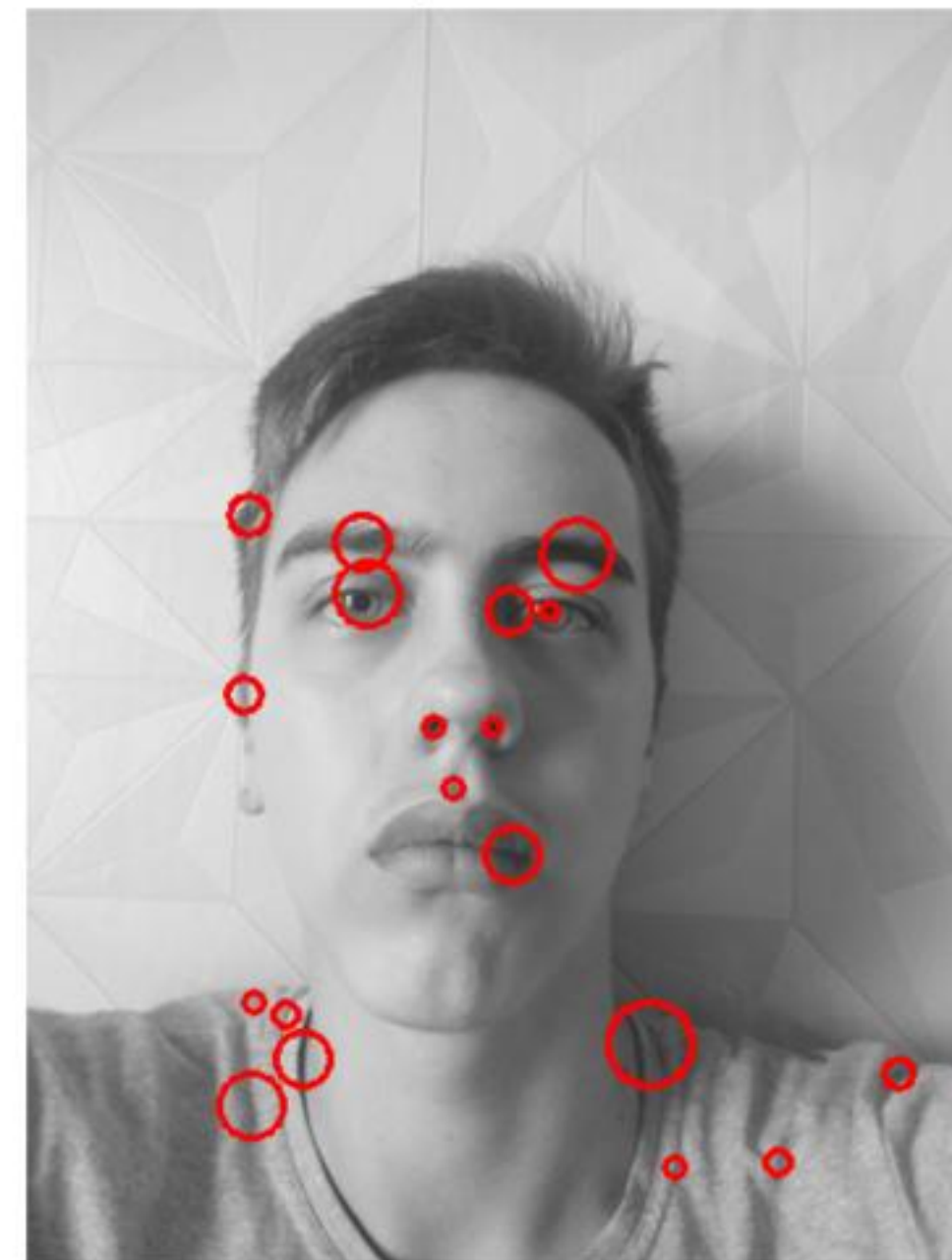
Person 9, Neutral: 30 blobs



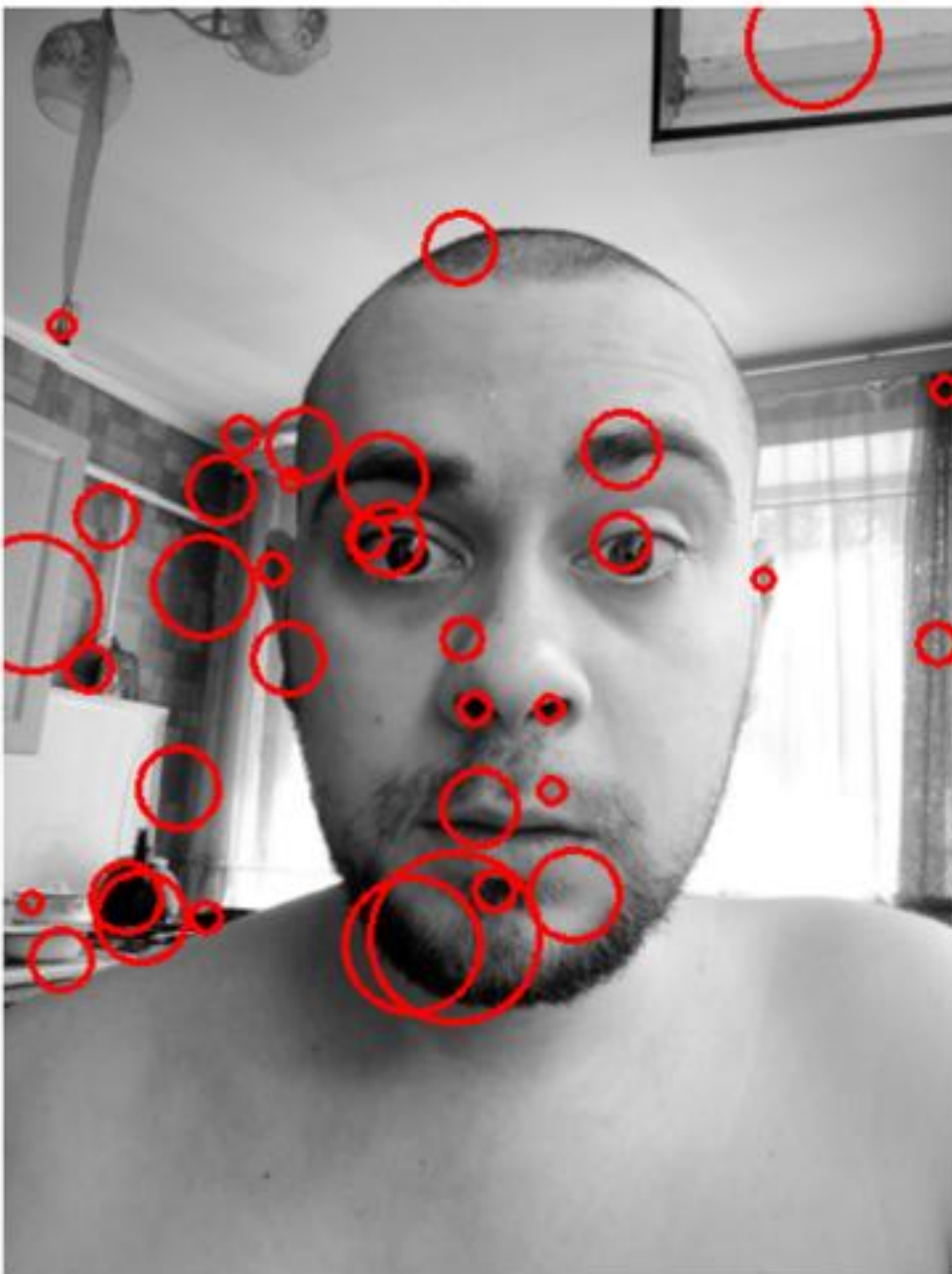
Person 13, Neutral: 32 blobs



Person 18, Neutral: 19 blobs



Person 3, Surprised: 36 blobs



Person 9, Surprised: 40 blobs



Person 13, Surprised: 51 blobs



Person 18, Surprised: 16 blobs



Dark blobs: 38 blobs



Bright blobs: 19 blobs



Dark blobs: 40 blobs



Bright blobs: 21 blobs



minArea=10: 82 blobs



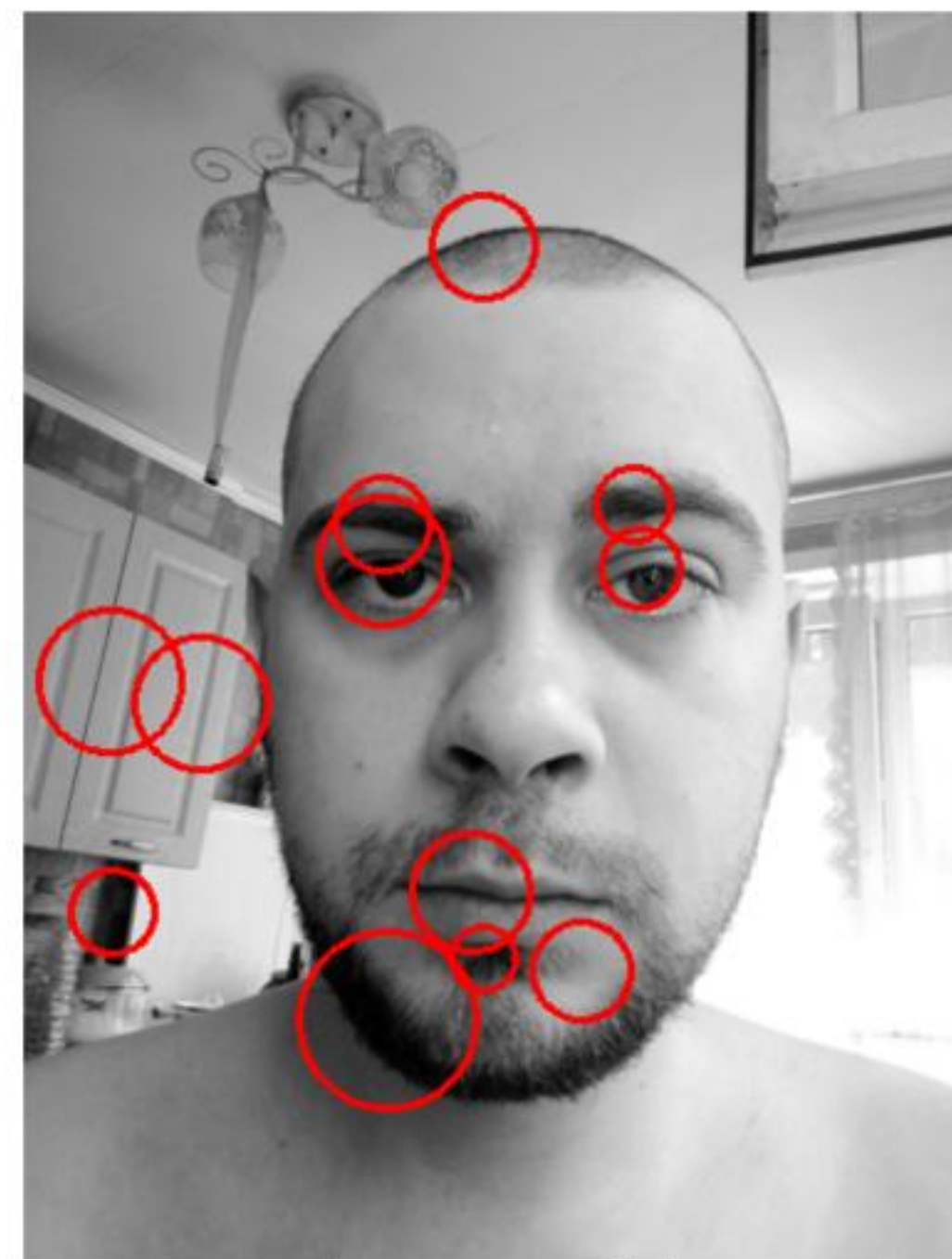
minArea=30: 38 blobs



minArea=100: 18 blobs



minArea=300: 12 blobs



minArea=10: 106 blobs



minArea=30: 40 blobs



minArea=100: 21 blobs



minArea=300: 9 blobs



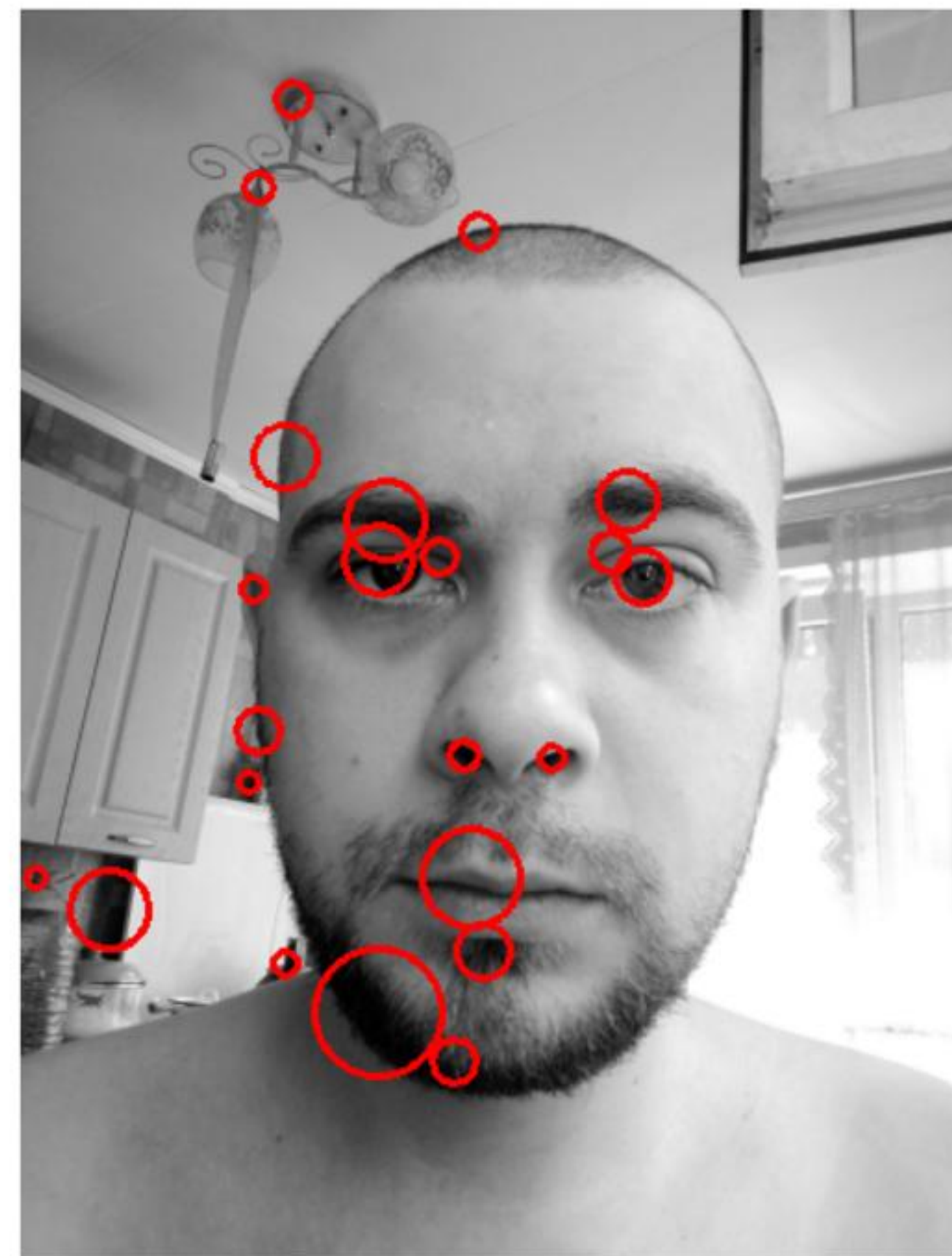
Thresholds 10-200: 38 blobs



Thresholds 50-150: 27 blobs



Thresholds 10-100: 22 blobs



Thresholds 10-200: 40 blobs



Thresholds 50-150: 29 blobs



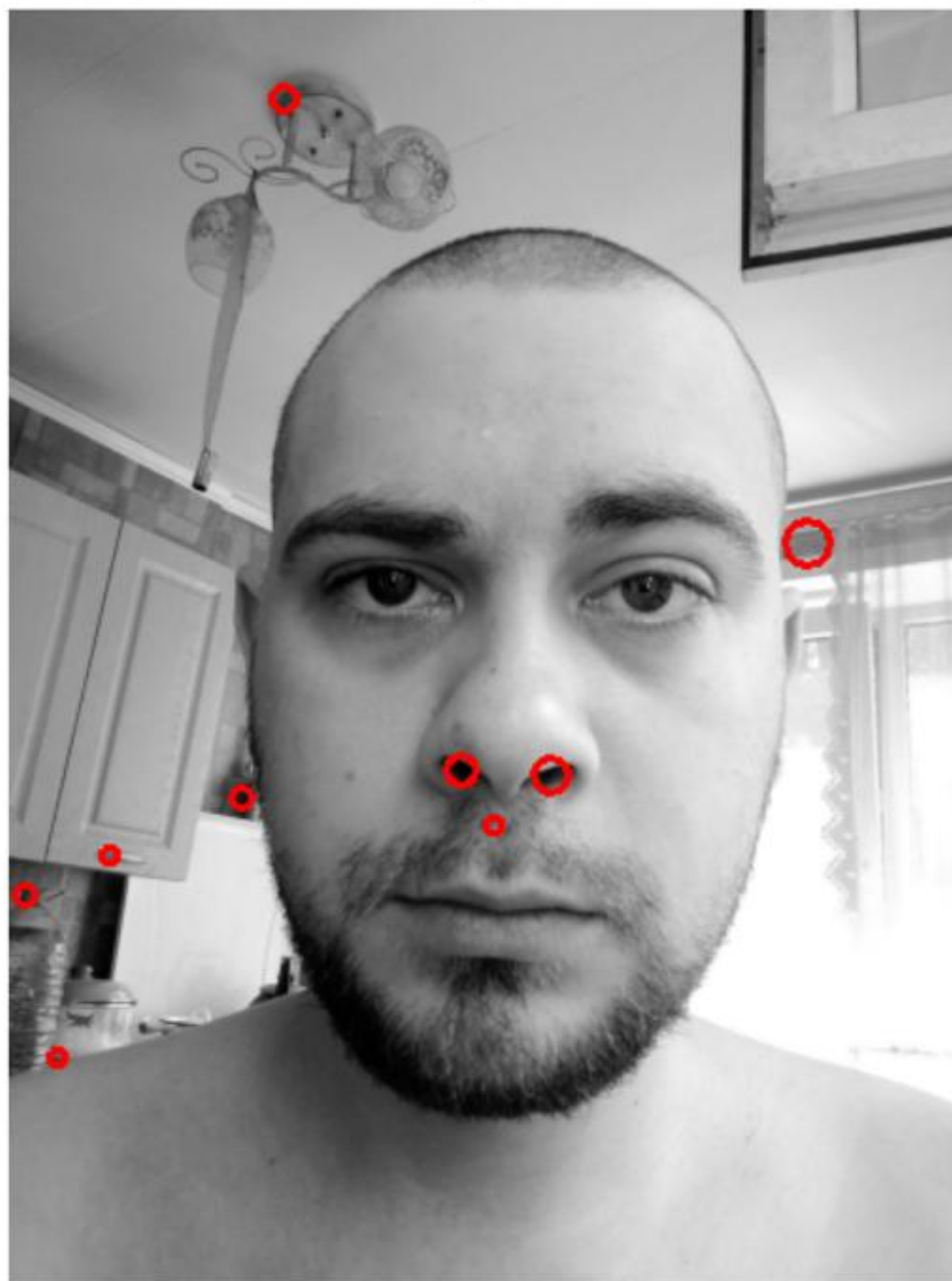
Thresholds 10-100: 23 blobs



No circularity filter: 38 blobs



minCircularity=0.5: 9 blobs



minCircularity=0.8: 0 blobs



No circularity filter: 40 blobs



minCircularity=0.5: 10 blobs



minCircularity=0.8: 2 blobs



Person 3, Neutral: 12 contours



Person 9, Neutral: 5 contours



Person 13, Neutral: 5 contours



Person 18, Neutral: 11 contours



Person 3, Surprised: 12 contours



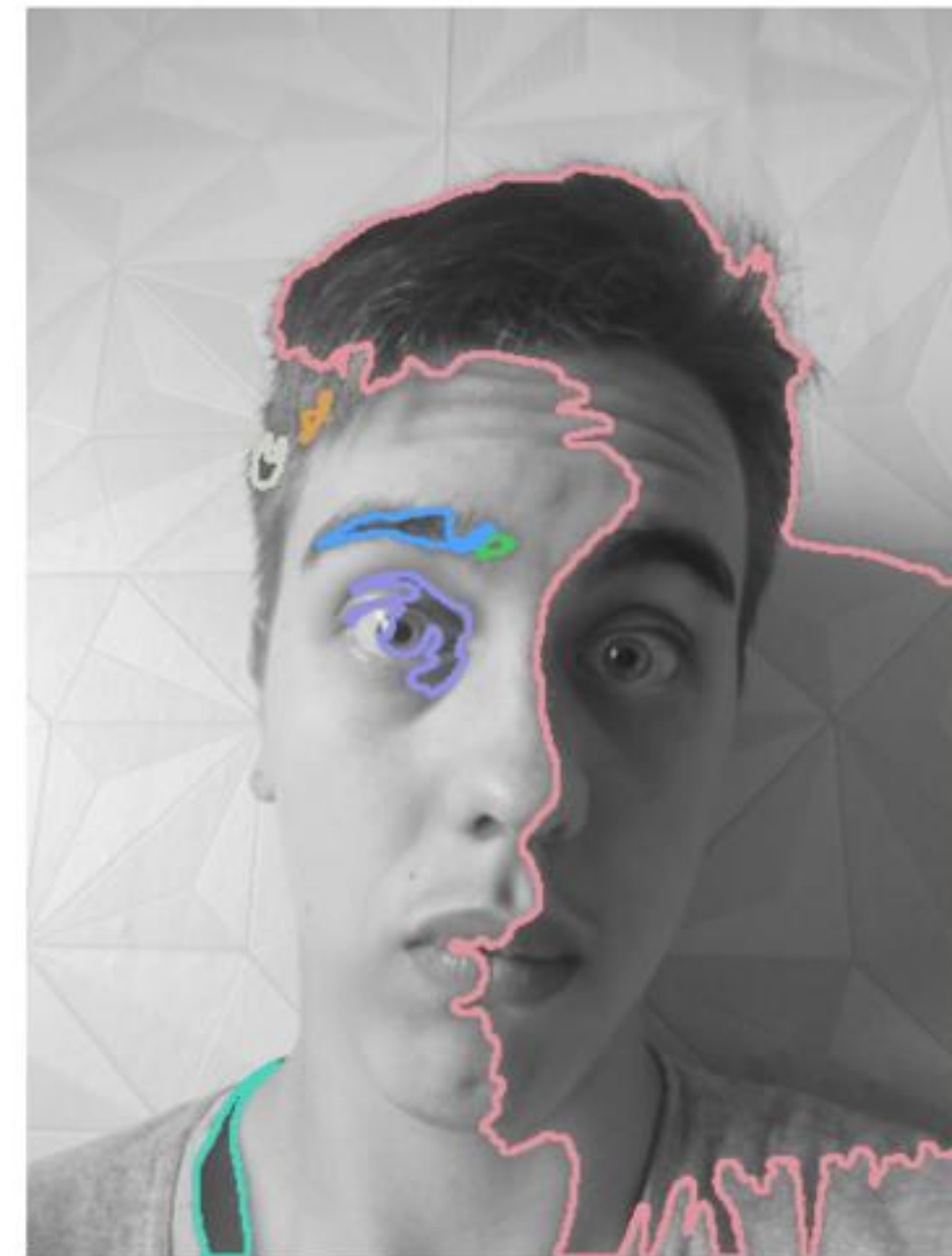
Person 9, Surprised: 3 contours



Person 13, Surprised: 8 contours



Person 18, Surprised: 7 contours



Blurred (5x5)



Binary (inverted threshold 120)



12 contours



Threshold 80: 13 contours



Threshold 120: 12 contours



Threshold 160: 7 contours



Otsu: 5 contours



Threshold 80: 13 contours



Threshold 120: 3 contours



Threshold 160: 5 contours



Otsu: 8 contours



Blur 3x3: 13 contours



Blur 7x7: 11 contours



Blur 11x11: 13 contours



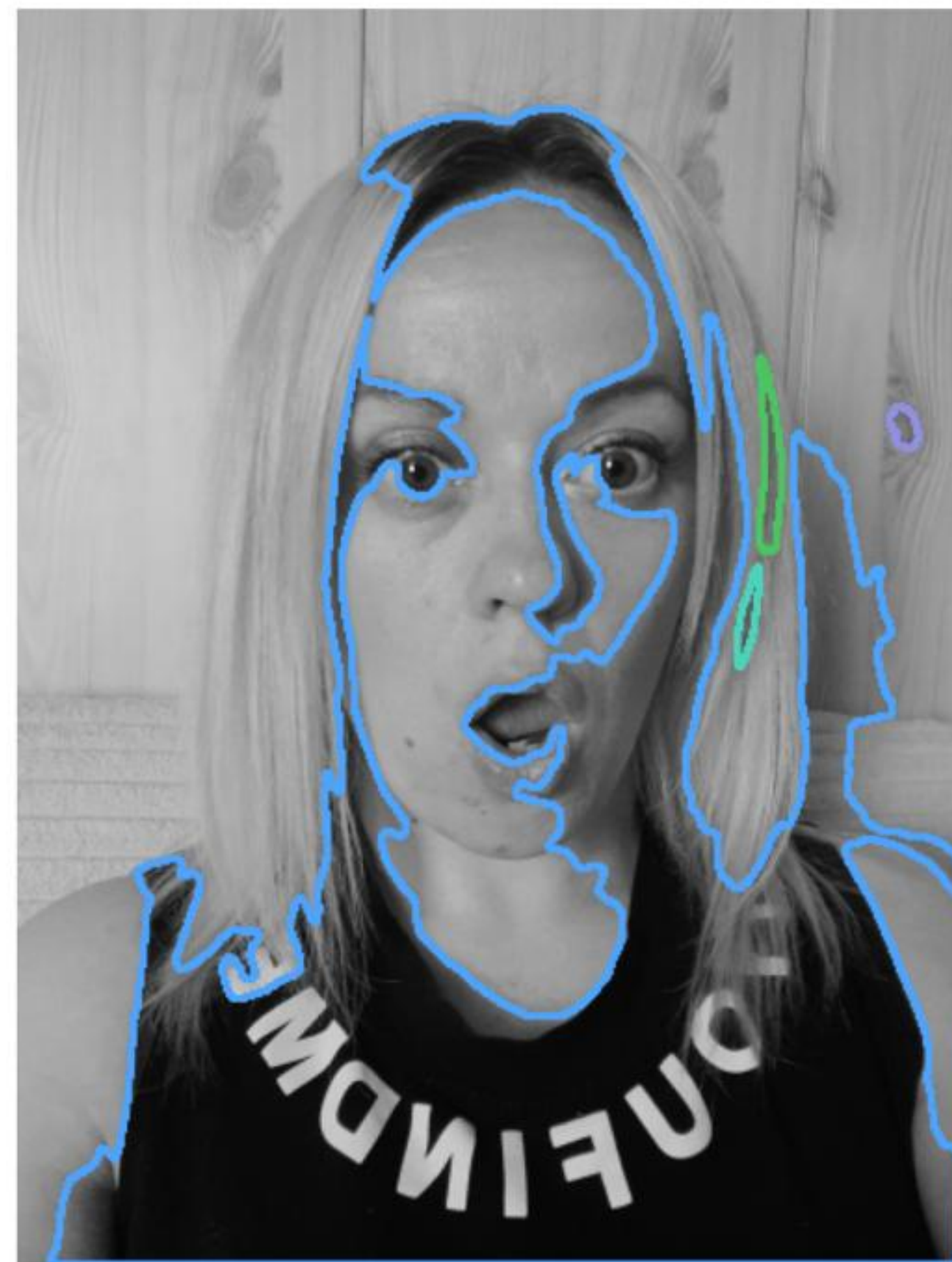
Blur 3x3: 3 contours



Blur 7x7: 3 contours



Blur 11x11: 4 contours



RETR_EXTERNAL: 12 contours



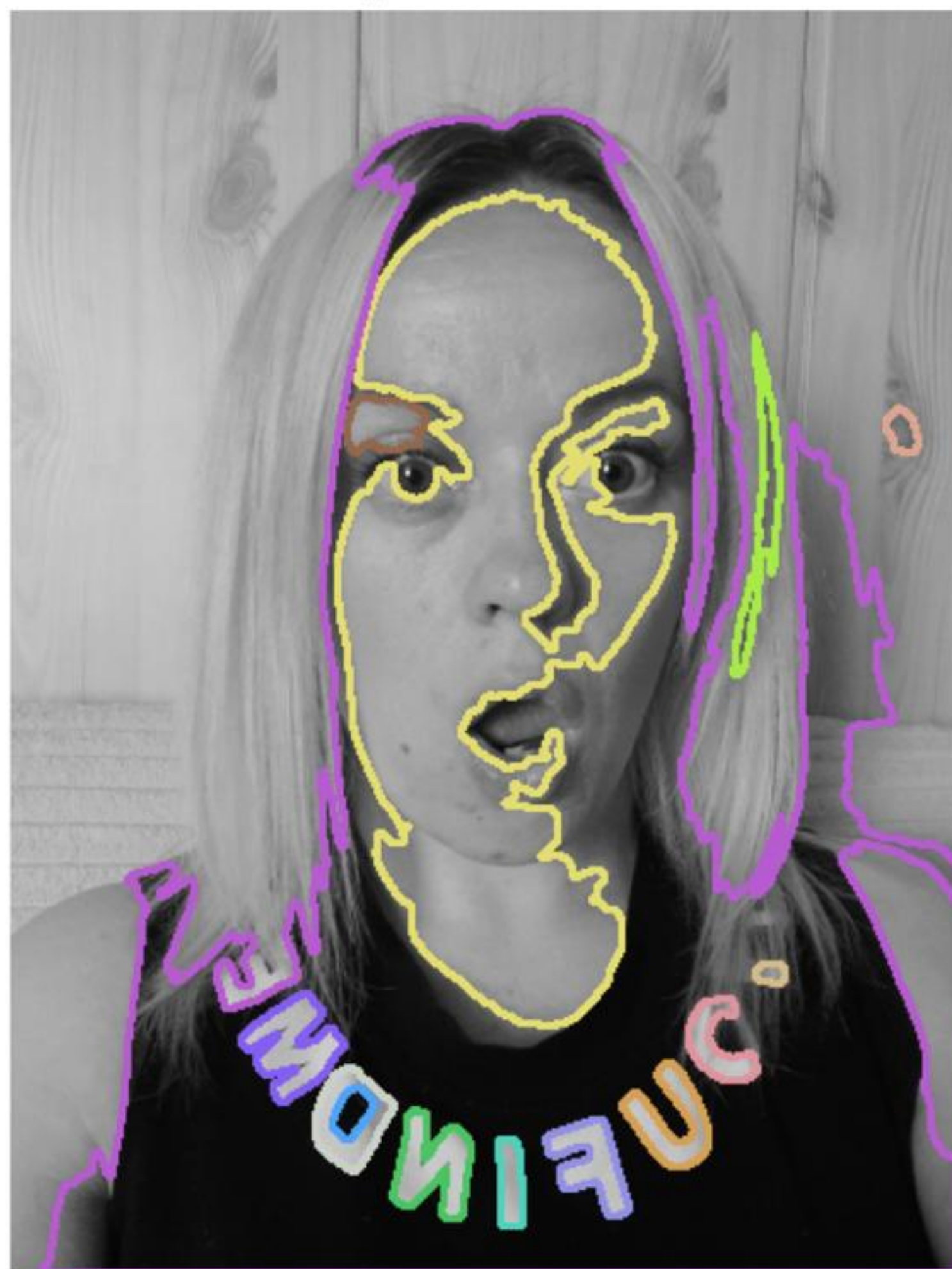
RETR_LIST: 13 contours



RETR_EXTERNAL: 3 contours



RETR_LIST: 14 contours



Dark regions (INV): 12 contours



Bright regions: 2 contours



Dark regions (INV): 3 contours



Bright regions: 12 contours



Canny 50/150: 70 contours



Canny 100/200: 43 contours



Canny 50/150: 57 contours



Canny 100/200: 39 contours



Blob detection: 38 blobs



Contour detection: 12 contours



Blob detection: 40 blobs



Contour detection: 3 contours

